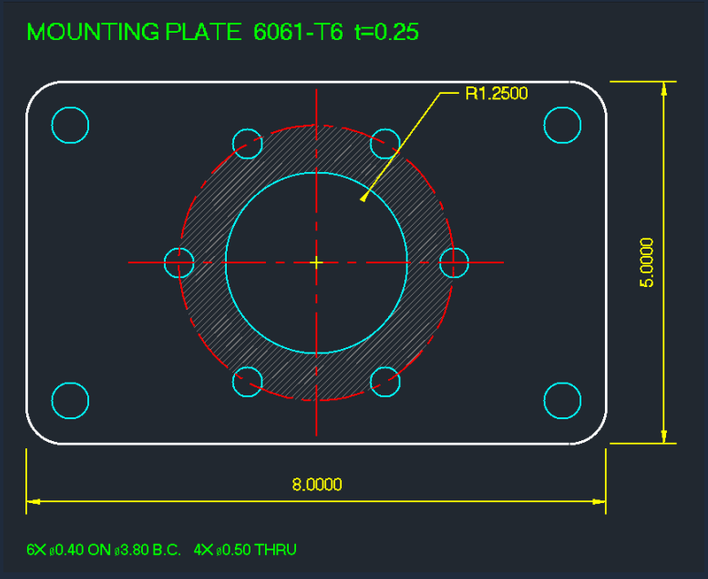
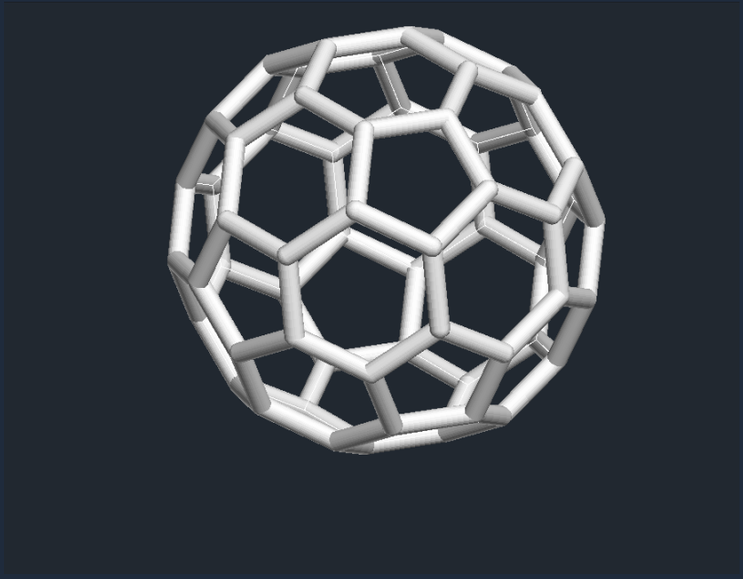
Fancy Pencil

User Manual

Drafting with AutoCAD-style commands and AutoLISP, in 2D and 3D, on Linux.

Version 0.5.0





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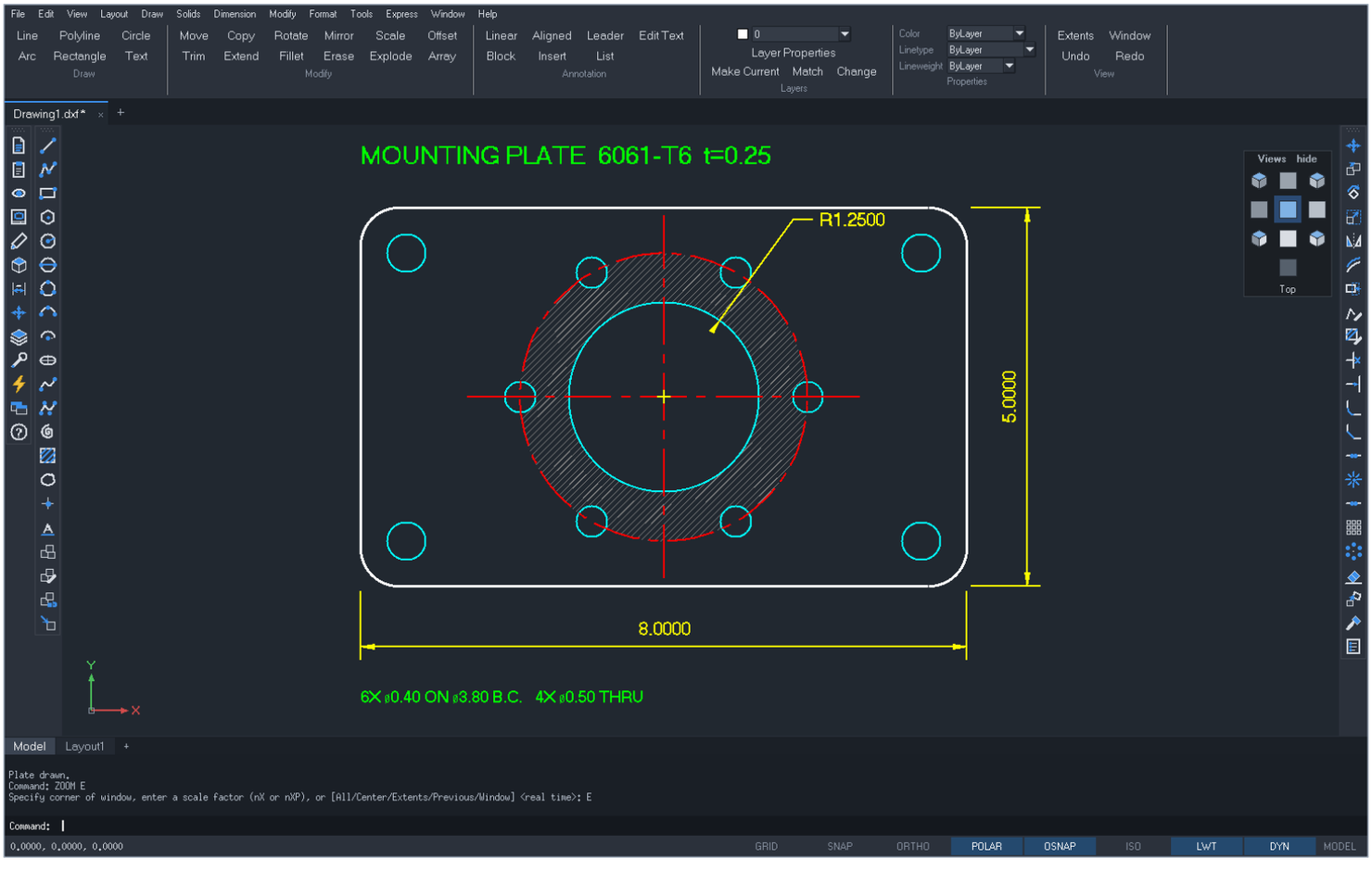
CHAPTER 1

Welcome to Fancy Pencil

A drafting program that works like AutoCAD: the same commands, the same prompts, and your AutoLISP.

Fancy Pencil is a 2D and 3D drafting program for Linux. If you have used AutoCAD, you already know how to use it: you type `LINE` or `L`, you answer the same prompts, you pick points with object snaps, you select objects first or after, and you save drawings that AutoCAD opens. Layers, blocks, dimensions, hatches, text styles, layouts and plotting all work the way you expect.

It also runs **AutoLISP**. Routines written for AutoCAD - `defun c:...`, `command`, `entget`, `ssget` filters, `getkword` and `*error*` handlers - run unchanged in most cases, and you can type AutoLISP straight at the command line.



Fancy Pencil with a drawing made by an AutoLISP program (the demo plate).

What it can do

- **Draw** lines, polylines, rectangles, polygons, circles, arcs, ellipses, splines, helixes, points, revision clouds and hatches.
- **Edit** with Move, Copy, Rotate, Scale, Mirror, Offset, Trim, Extend, Fillet, Chamfer, Array, Stretch, Align, Explode and grips.
- **Annotate** with single-line and paragraph text, dimensions, leaders and multileaders, in text and dimension styles.
- **Organise** with layers, linetypes, lineweights, colors, blocks and draw order.
- **Model in 3D**: boxes, cylinders, spheres and more; extrude, revolve, sweep, loft, press-pull; union, subtract, intersect and slice; the Platonic solids and wireframe struts.
- **Look around** in Top, Front, Side and isometric views, orbit, shade, and split the screen into viewports.
- **Print** to PDF or any printer from the model or from layouts with viewports, at true scale.
- **Exchange** drawings with AutoCAD as DXF, and as DWG with the free ODA File Converter.

How to read this manual

Chapters 2 to 4 get you started: installing, the window, and how commands are given. The chapters after them each cover one kind of work. Chapter 19 lists the keyboard shortcuts, and the Command Reference at the end lists every command with its aliases.

This manual is also inside Fancy Pencil: press `F1` or choose **Help > User Manual** to open it on screen.

Things you type are shown like `this`. A command's alias (its short name) is shown in brackets: LINE (`L`). Menu paths are written with arrows: **Format > Layer**. Keys are written `Ctrl+S`, `F3`.

CHAPTER 2

Installing Fancy Pencil

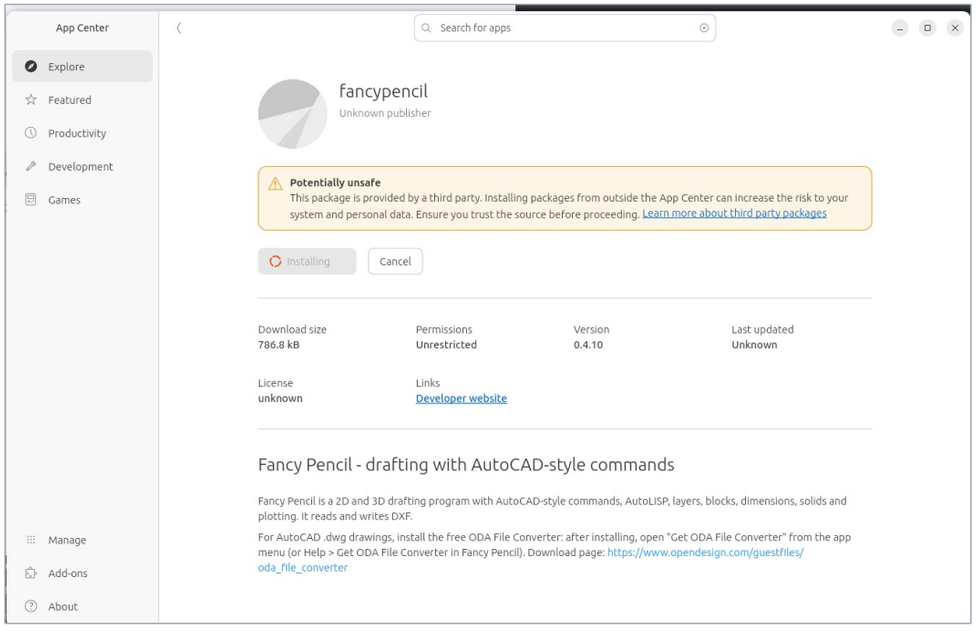
One file installs everything: Install Fancy Pencil.deb, from the Fancy Pencil Kit.

What you need

- A 64-bit computer with **Ubuntu** 22.04 or newer, **Linux Mint** 21 or newer, or **Debian** 12.
- The **Fancy Pencil Kit** folder (on a USB stick or drive, or downloaded). It holds `Install Fancy Pencil.deb` and `INSTRUCTIONS.txt`.
- Your password, to install software.

Install

1. Open the Fancy Pencil Kit folder and double-click **Install Fancy Pencil.deb**.
2. If it opens in Archive Manager instead (a list of folders called DEBIAN, opt and usr), close that window, right-click the file and choose **Open With... > App Center**.
3. App Center shows Fancy Pencil. Click **Install** and enter your password. App Center also installs SBCL and Tk, the two programs Fancy Pencil runs on.
4. When it has finished, press the Super (Windows) key, type **Fancy Pencil**, and click its icon. Right-click the icon in the dock and choose **Pin to Dash** to keep it there.



App Center installing Fancy Pencil from the Kit.

> **Note** App Center shows **Potentially unsafe** and **Unknown publisher** for every program that comes from a file instead of its own store. It is not a fault in Fancy Pencil. Install it if you trust where the Kit came from.

If App Center won't open the file

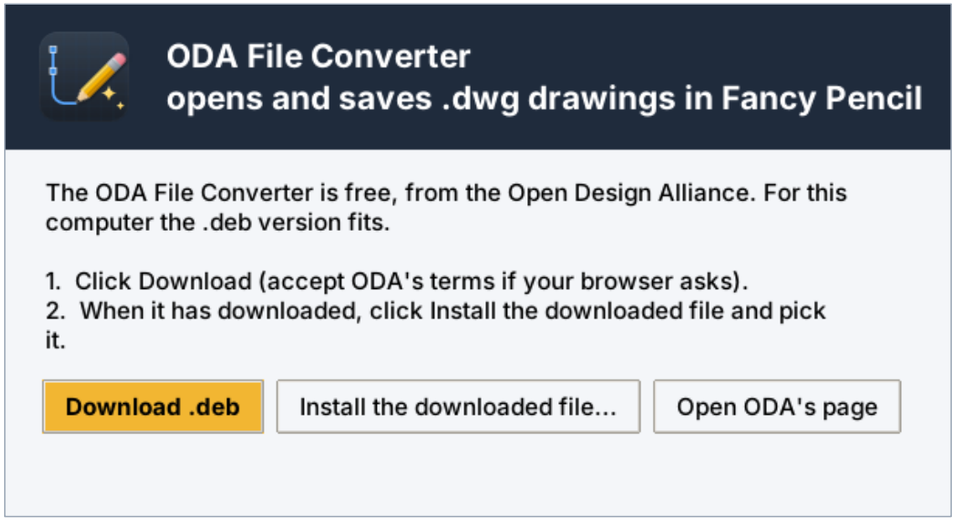
Right-click an empty part of the Kit folder, choose **Open in Terminal**, and type:

```
sudo apt install "./Install Fancy Pencil.deb"
```

AutoCAD DWG files: ODA File Converter

Fancy Pencil reads and writes AutoCAD's DXF files by itself. For **.dwg** files it uses the free **ODA File Converter** from the Open Design Alliance. To install it:

1. Open **Get ODA File Converter** from the app menu (or, in Fancy Pencil, **Help > Get ODA File Converter**).
2. Click **Download .deb**. Your web browser downloads the converter; accept ODA's terms if it asks.
3. When the download has finished, click **Install the downloaded file...**, pick the file whose name starts with `ODAFileConverter` and ends in `.deb`, and enter your password.



The Get ODA File Converter window.

Download page: `www.opendesign.com/guestfiles/oda_file_converter`. Fancy Pencil finds the converter by itself once it is installed.

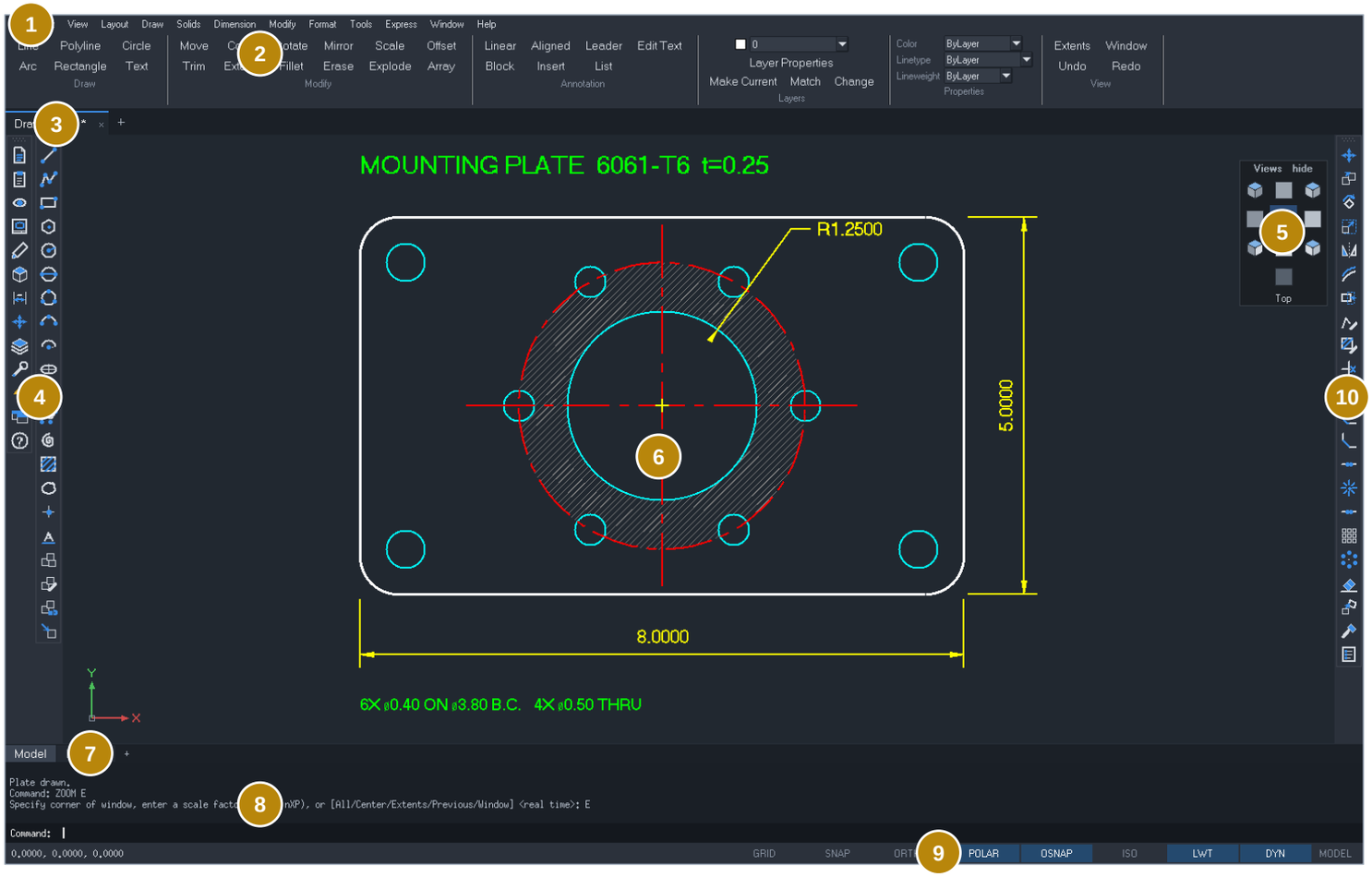
Updating and removing

- **Update**: double-click a newer Install Fancy Pencil.deb and click **Install** (or **Update**). Your settings, toolbars and drawings are kept.
- **Remove**: in App Center, **Manage**, find fancypencil and remove it; or type `sudo apt remove fancypencil`.
- **Your settings** live in the hidden folder `~/.config/fancypencil`: preferences, recent files, aliases, toolbars and plot settings. Copy it to another computer to bring them along.

CHAPTER 3

The Fancy Pencil Window

Where everything is.



The parts of the window.

1	**Menu bar**	Every command, by subject: File, Edit, View, Layout, Draw, Solids, Dimension, Modify, Format, Tools, Express, Window, Help.
2	**Ribbon**	The everyday commands in groups (Draw, Modify, Annotation, Layers, Properties, View), the current layer list and the color, linetype and lineweight lists.
3	**Drawing tabs**	One tab for each open drawing. Click to switch, x to close, + for a new drawing.
4	**Toolbars**	Icons for commands. The Menus toolbar opens any menu's whole tree; Draw is beside it. Rest the mouse on an icon to see its name.
5	**View picker**	Click a cube to see the drawing from the Top, a side, or an isometric corner.

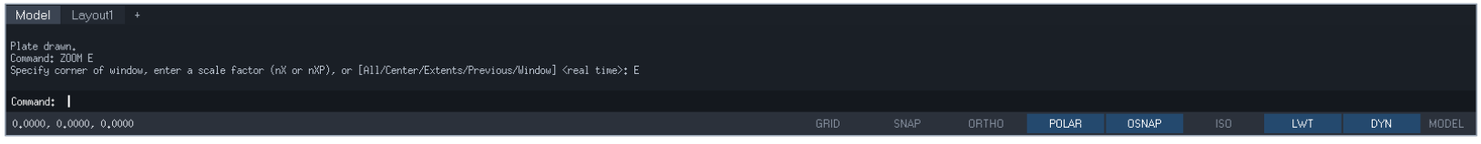
6	**Drawing area**	Where you draw. Wheel zooms at the cursor, middle-drag pans, middle double-click shows everything.
7	**Model and layout tabs**	Model is where you draw full size; layouts are sheets of paper for plotting.
8	**Command line**	Where commands are typed and prompts appear. The history above it shows what happened.
9	**Status bar**	The cursor's coordinates, and buttons for GRID, SNAP, ORTHO, POLAR, OSNAP, ISO, LWT and DYN.
10	**Modify toolbar**	Move, Copy, Rotate, Scale, Mirror, Trim, Extend, Fillet, Array and the other editing commands.

The ribbon



The ribbon: click a word to run that command.

Command line and status bar



Prompts appear here; the status bar buttons turn drawing aids on and off.

The status bar buttons

GRID	`F7`	A grid of dots or lines to help you judge sizes. It doesn't print.
SNAP	`F9`	The cursor jumps in steps (the snap spacing).
ORTHO	`F8`	Lines go only straight across or straight up and down.
POLAR	`F10`	Guides at set angles while you draw.
OSNAP	`F3`	Object snaps: the cursor locks onto ends, middles, centres and more.
ISO	`F5`	Isometric drafting, and which isoplane is current.
LWT		Show lineweights (line thickness) on screen.
DYN	`F12`	Show prompts and values next to the cursor.
MODEL		Shows Model or Paper space; on a layout, click to switch.

Several drawings at once

Every drawing opens in its own tab. `Ctrl+Tab` and `Ctrl+Shift+Tab` step through them, `Ctrl+F4` or `CLOSE` closes the current one, and the **Window** menu lists them all. Each drawing keeps its own view, selection, undo history, layers and settings. Copy (`Ctrl+C`) and Paste (`Ctrl+V`) work between drawings, bringing layers, styles and blocks along.

When you close a drawing its view is remembered; the next time you open it, it comes back zoomed where you left it.

CHAPTER 4

Giving Commands

The command line, coordinates, and drawing aids.

Typing commands

- Type a command or its alias and press `Enter` or `Space`: `LINE` or `L`, `CIRCLE` or `C`, `TRIM` or `TR`.
- Answer the prompts the same way. Options are shown in square brackets: `[Close/Undo]`. Type the capital letters of one (`C`, `U`) to choose it, or click it. Or press `Down` (the arrow key) for a menu of them at the cursor; pick one with the arrow keys and `Enter`, or click it.
- The value in angle brackets is the default: `<1.0000>`. Press `Enter` to accept it.
- `Enter` on an empty prompt repeats the last command. `Esc` cancels. While a command runs, right-click is `Enter`.
- Menus, the ribbon and toolbar icons type the command for you; the prompts are the same.

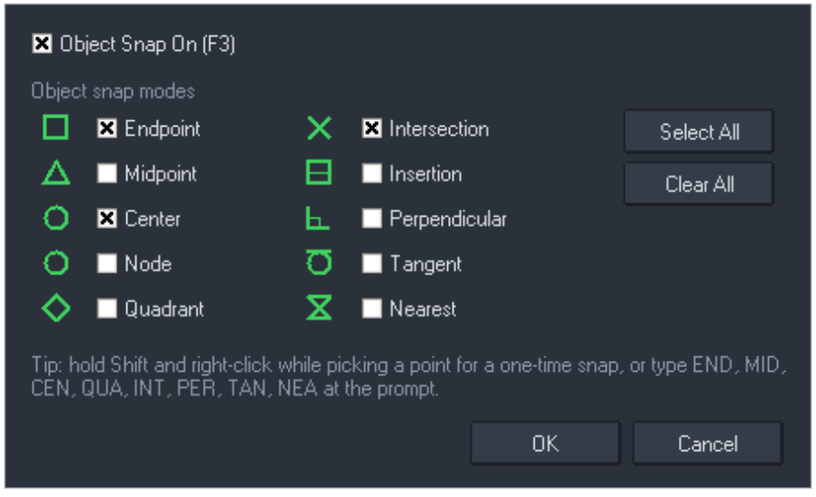
Typing points and distances

`3,4`	The point at X 3, Y 4 (the first point of a command).
`@2,0`	2 to the right of the last point (relative).
`@5<45`	5 away from the last point, at 45 degrees (polar).
`4,0`	After the first point, typed points are relative to the last one, as with AutoCAD's dynamic input: LINE `0,0` then `4,0` then `0,3` goes 4 across and 3 up.
`#10,10`	An absolute point, even after the first.
`5`	A bare number: 5 along the rubber band toward the cursor (direct distance).
`2'6 1 1/2`	Feet and inches, and fractions, wherever a distance is asked for.

`SETVAR DYNPICOORDS 1` makes every typed point absolute (remembered). Points given by AutoLISP and scripts are always absolute.

Object snaps

Object snaps put the point exactly on an object: its end, its middle, a circle's centre, where two lines cross. With **OSNAP** on (`F3`), the snap markers appear as the cursor nears an object; click to take the point. `OSNAP` (or right-click the OSNAP button) chooses which snaps are on.



Object Snap settings (OSNAP).

For one point only, type the snap's short name at the prompt: `END`, `MID`, `CEN`, `INT`, `PER`, `TAN`, `QUA`, `NOD`, `INS`, `NEA` - or `NON` for none. **Shift + right-click** gives a menu of them.

Snaps work on solids in every view, plan or 3D: the corners, edge middles and round edges' centres, and also their **faces**. **CEN** over a flat face snaps to the middle of that face; **NEA** with no edge near puts the point on the face under the cursor, so you can draw right onto a solid. In the 3D views **INT** finds where edges really meet in space, and where a line passes through a face; **PER** finds the foot of the perpendicular anywhere along the edge under the cursor.

Ortho, Polar, Grid and Snap

`F8` ORTHO keeps lines straight across or up and down. `F7` GRID shows a grid and `F9` SNAP makes the cursor jump in steps. In the isometric views Ortho follows the X and Y of the plane you are drawing on; in Front, Back, Left and Right it follows the screen, so up the screen is up in Z.

AutoLISP at the command line

Type AutoLISP straight in: `(+ 1 2)`, `!myvar` to see a variable, or a longer expression over several lines (the prompt shows `(_>` until the brackets balance). Expressions also work inside prompts: at "Specify next point" you can type `(polar (getvar "lastpoint") 0 5)`.

Undo and redo

`U` or `Ctrl+Z` undoes the last command, `REDO` or `Ctrl+Y` brings it back, and `UNDO` undoes several at once. Each drawing has its own undo history.

CHAPTER 5

Selecting, Grips and Properties

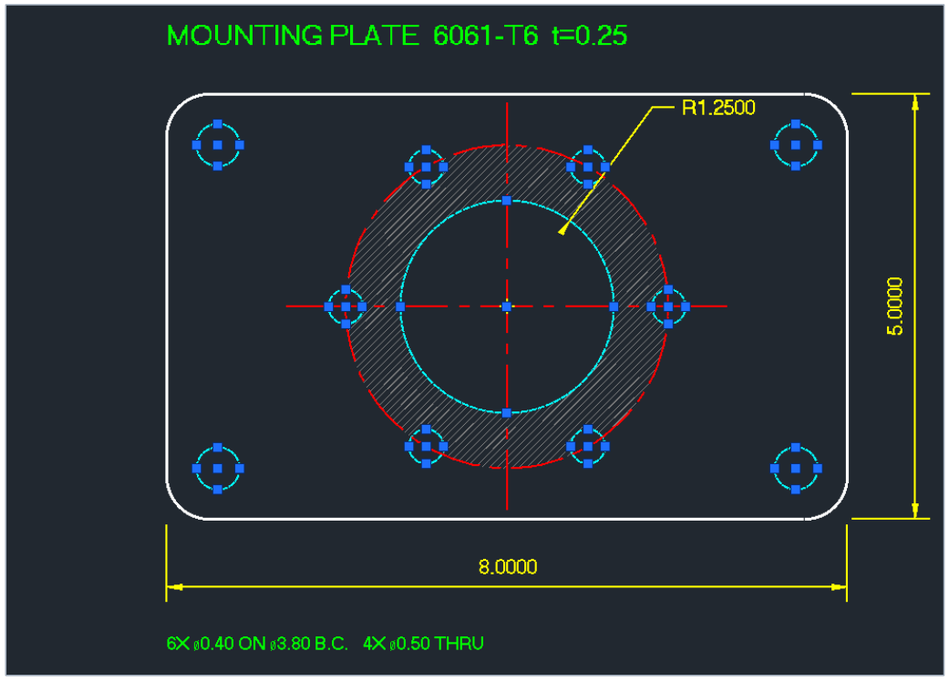
Pick objects, change them by dragging, and read or edit everything about them.

Selecting

- **Click** an object at the `Command:` prompt to select it. Click more objects to add them.
- **Drag left to right** for a window: it selects what is completely inside.
- **Drag right to left** for a crossing window: it selects what is inside or touches it.
- `Ctrl+A` (SELECTALL) selects everything. `Esc` clears the selection.
- Then run a command - MOVE, ERASE (or press `Delete`) - and it uses the selection. Or start the command first and select when it asks.

Grips

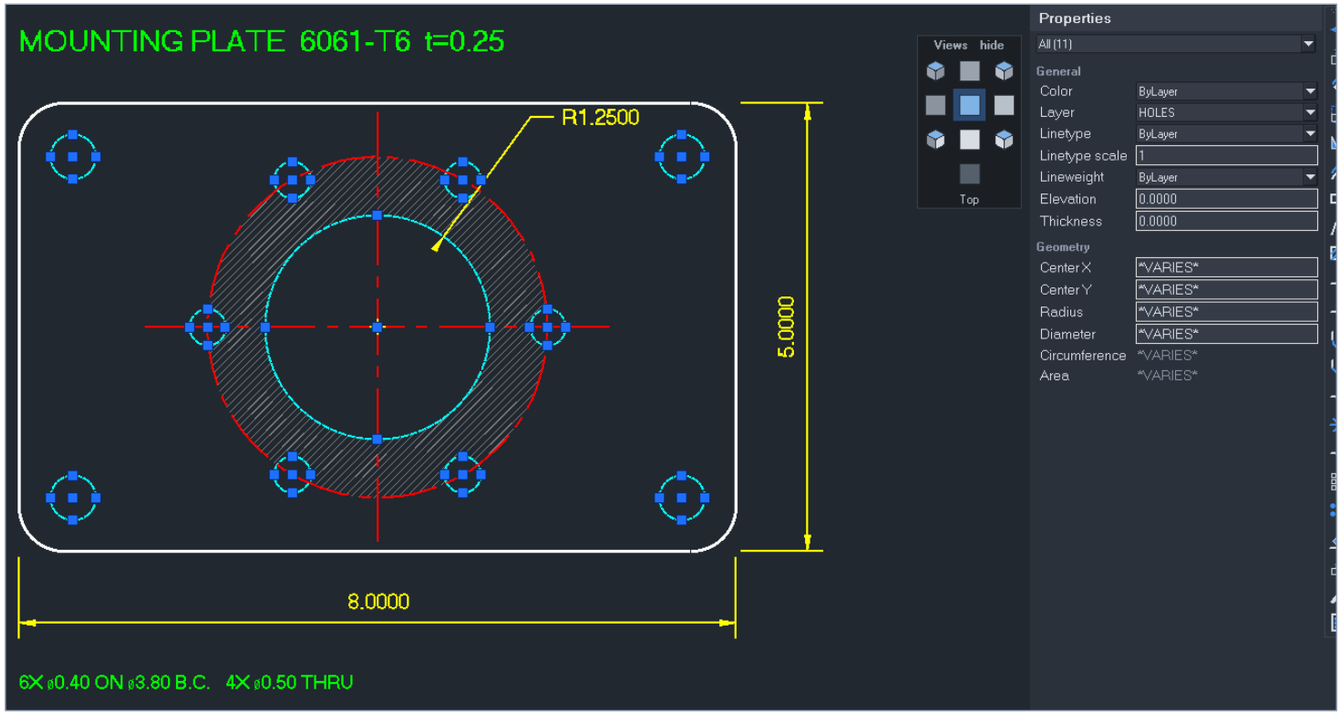
Selected objects show blue squares - **grips** - at their ends, middles and centres. Click a grip to make it hot (red) and drag it: a line's end moves, a circle grows. With a hot grip, `Enter` or `Space` steps through STRETCH, MOVE, ROTATE, SCALE and MIRROR, with the Base point, Copy, Undo and eXit options. Grips on several objects that share a point move them all together.



Six bolt holes selected: their grips show at the centres and quadrants.

The Properties palette

`PROPERTIES` (`Ctrl+1`) opens the palette on the right. With nothing selected it shows the settings new objects will get (color, layer, linetype, lineweight). With a selection it shows what the objects share, and for one kind of object its sizes too - a circle's centre, radius, diameter, circumference and area. Change a value and press `Enter`. The list at the top ("All (6)", "Circle (6)") narrows a mixed selection to one kind.



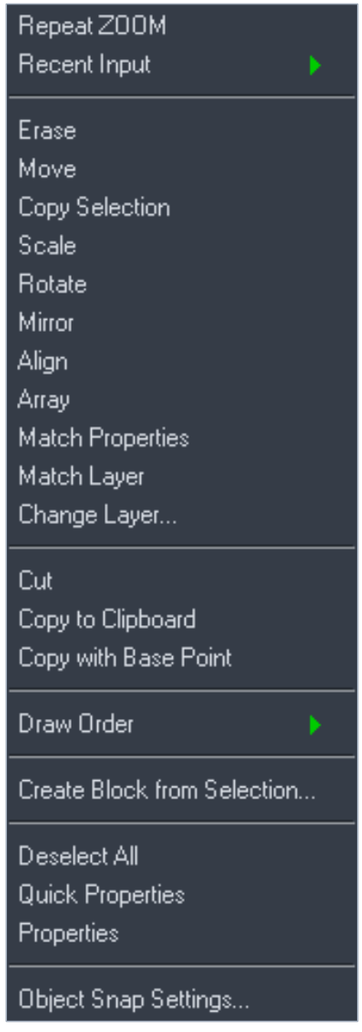
The Properties palette showing the six selected circles.

Quick Properties and rollover tips

Quick Properties (`QP`, or right-click > Quick Properties) is a small panel with just the main properties; it stays open and follows whatever you select. Rest the cursor on any object for half a second and a **rollover tip** shows what it is and its sizes.

The right-click menu

Right-click the drawing at the Command prompt: you get Repeat, Recent Input and, with objects selected, Erase, Move, Copy, Scale, Rotate, Mirror, Align, Array, Draw Order, Solid Editing (Union, Subtract, Intersect, Slice) for solids, Create Block, Match Properties and Properties.



Right-click with objects selected.

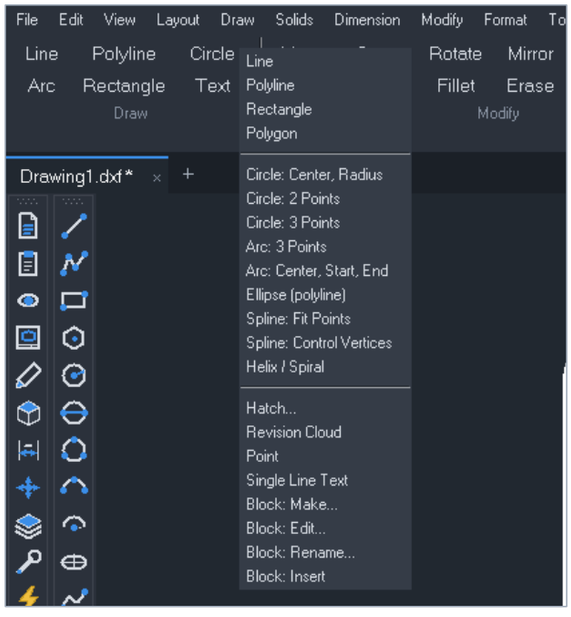
Measuring

`DIST` (`DI`)	Distance and angle between two points.
`ID`	The coordinates of a point.
`AREA` (`AA`)	Area and perimeter of points you pick or of an object.
`LIST` (`LI`)	Everything about the selected objects, in a window you can copy from.
`MASSPROP`	Area, centroid and moments of closed shapes; volume and mass of solids.

CHAPTER 6

Drawing

Lines, curves, shapes and fills.



The Draw menu.

Lines and polylines

- **LINE** (`L`): click points; each pair is a separate line. `C` closes back to the start, `U` takes back a point, `Enter` ends.
- **PLINE** (`PL`): one object made of straight and curved segments. `A` switches to arcs, `L` back to lines, `W` sets a width.
- **RECTANG** (`REC`): two corners. Typing `11,8.5` after the first click makes an 11 x 8.5 rectangle.
- **POLYGON** (`POL`): number of sides, the centre, inscribed or circumscribed, and the size.

Circles, arcs and ellipses

- **CIRCLE** (`C`): centre and radius, or `D` for a diameter; `2P`, `3P` and Tan-Tan-Radius.
- **ARC** (`A`): three points, or Start-Center-End and the other AutoCAD ways.
- **ELLIPSE** (`EL`): axis endpoints, or the centre; `Isocircle` in isometric drafting.

Splines and helixes

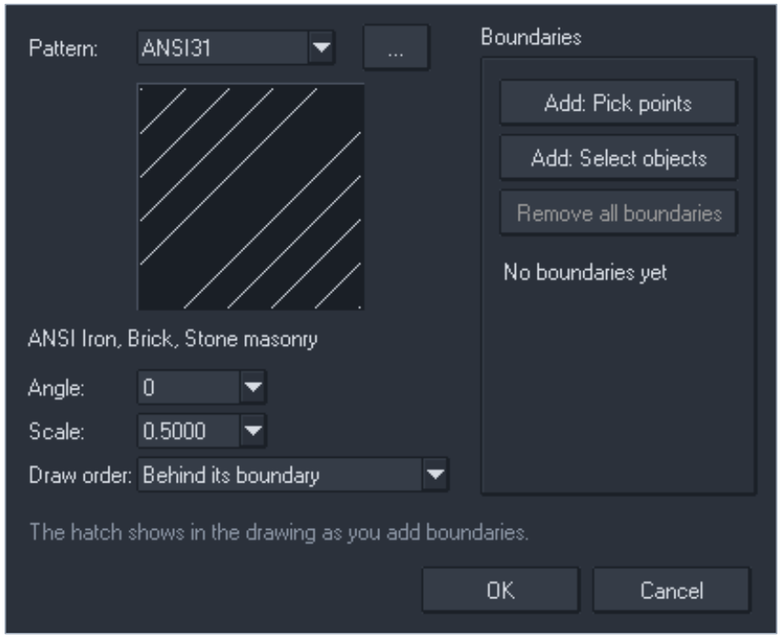
SPLINE (`SPL`): pick points and press `Enter`. With **Method Fit** the curve passes through every point; **Method CV** treats them as control vertices that pull the curve. Close, start and end Tangency, and Knots shape it. **HELIX** makes springs, thread paths and flat spirals: base centre, base and top radius, then height, with Turns, turn Height and tWist.

Points, revision clouds and text

POINT (`PO`) places a point in the style set by **DDPTYPE** (Format > Point Style). **REVCLOUD** (`RC`) draws a cloud around changes. Text has its own chapter.

Hatching

HATCH (`H`) fills an area. In the dialog, choose a pattern (or click the swatch for the palette), the angle and the scale, then **Add: Pick points** and click inside the area: the boundary is found automatically, with closed objects inside left as islands. The hatch shows as you go; `Enter` brings the dialog back, and **OK** keeps it.



The Hatch dialog.

Patterns include SOLID, ANSI31 to ANSI38, LINE, NET, DASH, DOTS, BRICK, SQUARE, CROSS, HONEY, EARTH, STEEL, INSUL and more. **HATCHEDIT** (`HE`) changes an existing hatch; **-HATCH** asks at the command line, for scripts.

CHAPTER 7

Modifying

Move, copy, trim, round off and rearrange.

Moving and copying

MOVE (`M`)	Select, a base point, then where it goes.
COPY (`CO`)	The same, leaving the original; keep clicking for more copies.
ROTATE (`RO`)	A base point and an angle, or Reference to line one thing up with another.
SCALE (`SC`)	A base point and a factor, or Reference: "this length should be that length".
MIRROR (`MI`)	Two points on the mirror line; keep or erase the originals.
ARRAY (`AR`)	Copies in rows and columns (Rectangular) or around a centre (Polar).
ALIGN (`AL`)	Pairs of points: the source points move onto the destination points, turning and optionally scaling. Works in 3D with up to three pairs.
STRETCH (`S`)	A crossing window over the part to stretch, then a base point and where it goes.

Trimming and extending

TRIM (`TR`): select the cutting edges (or press `Enter` for all), then click the parts to cut off. **EXTEND** (`EX`) is the same, lengthening objects to the edges. `Shift`+click switches between them. They work on lines, arcs, circles and polylines. **BREAK** (`BR`) cuts a gap or splits an object at a point.

Offset, fillet and chamfer

- **OFFSET** (`O`): a distance, then the object and the side - a parallel copy.
- **FILLET** (`F`): two lines (or a polyline with `P`) joined by an arc of the fillet radius (`R`). Radius 0 makes a sharp corner.
- **CHAMFER** (`CHA`): the same with a straight cut.
- **JOIN** (`J`) joins touching pieces into one; **EXPLODE** (`X`) breaks polylines, blocks, dimensions, text and solids into their parts; **PEDIT** (`PE`) edits polylines.

Changing properties

- **MATCHPROP** (`MA`): pick a source object, then the objects to make like it. `S` chooses what is copied: color, layer, linetype, lineweight, text and dimension style, hatch pattern...
- **CHPROP**, the Properties palette, and the ribbon's Color, Linetype and Lineweight lists change selected objects directly.

Draw order and text case

Later objects are drawn over earlier ones. **DRAWORDER** (`DR`) sends selected objects to the Front or Back, or Above or Under others; **TEXTTOFRONT** and **HATCHTOBACK** do it for all text or all hatches. **TCASE** changes text to UPPER, lower, Sentence, Title or tOGGLE case.

Erasing and getting things back

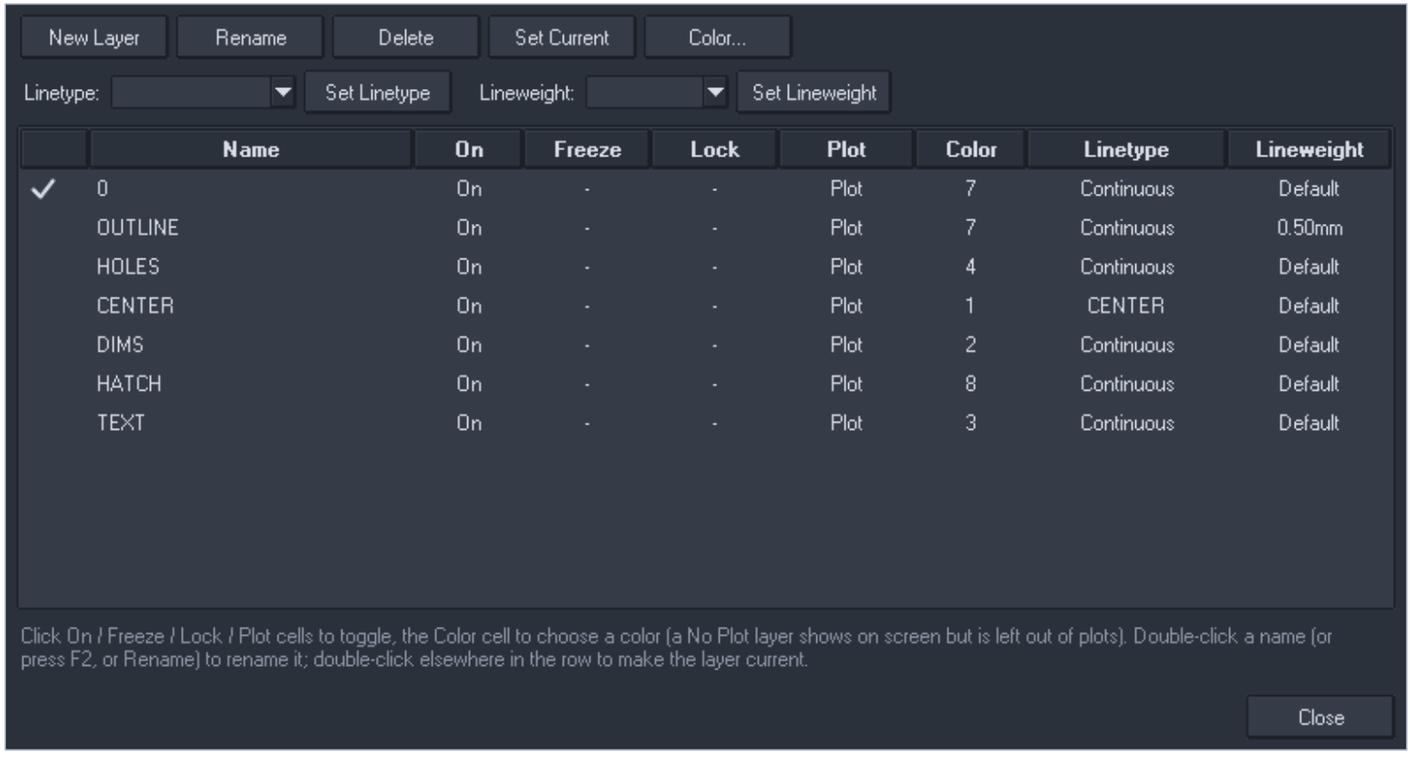
ERASE (`E`, or the `Delete` key) removes the selection; **OOPS** brings back the last erased objects; `U` undoes any command.

CHAPTER 8

Layers, Colors and Linetypes

Keep a drawing organised and make it print the way you want.

Layers are like sheets of clear film stacked on each other: walls on one, dimensions on another, notes on a third. Each layer has a color, linetype and lineweight that its objects use, and can be turned off, frozen or locked.



The Layer Properties Manager (LAYER).

The Layer Properties Manager

- Open it with **LAYER** (LA), **Format > Layer**, or Layer Properties on the ribbon.
- **New Layer** adds one with its name ready to type. Double-click a name (or F2) to rename it.
- Click **On**, **Freeze**, **Lock** or **Plot** in a row to switch it; click **Color**, **Linetype** or **Lineweight** to change them.
- Double-click a row (or **Set Current**) to make it the current layer - new objects go on it.
- A **No Plot** layer shows on screen but is left out of plots, like Defpoints.

Quick layer tools

Make Current (`LAYMCUR`)	Click an object: its layer becomes the current layer.
Match (`LAYMCH`)	Move the selected objects to the layer of an object you click.
Change (`CL`)	Move the selected objects to a layer picked from a list.
`LAYCUR`	Move the selected objects to the current layer.
`LAYISO` / `LAYUNISO`	Show only the layers of the selected objects, and back.
`LAYOFF` `LAYFRZ` `LAYLCK`	Turn off, freeze or lock an object's layer by clicking it (and LAYON, LAYTHW, LAYULK to undo).

Choosing a layer in the ribbon's layer list while objects are selected moves them to it; with nothing selected it sets the current layer.

Colors

Colors open the **Select Color** dialog: the 255 AutoCAD index colors, the standard colors and grays, ByLayer (the layer's color) and ByBlock. White objects (color 7) show black on a light background and print black.

Linetypes and lineweights

LINETYPE (`LT`) loads CENTER, HIDDEN, DASHED, PHANTOM, DOT, DASHDOT, BORDER, DIVIDE and their variants, or any `.lin` file. **LTSCALE** (`LTS`) sets the size of the dashes for the whole drawing. **LWEIGHT** (`LW`) sets line thickness; turn on **LWT** in the status bar to see it.

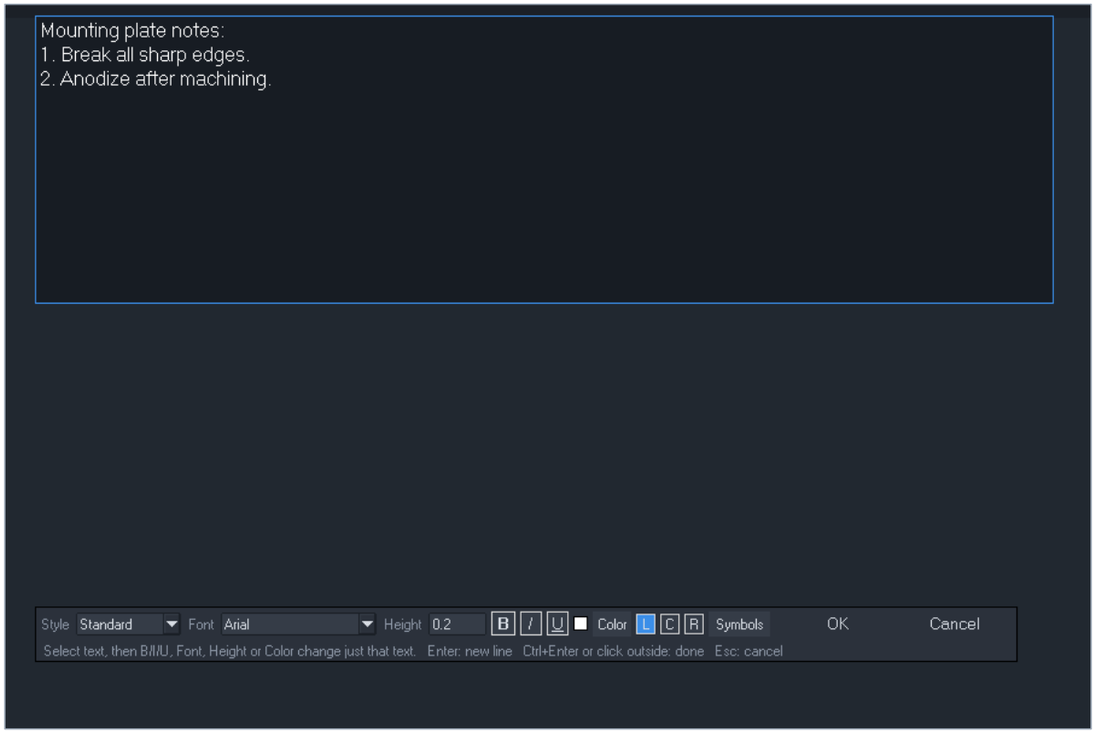
CHAPTER 9

Text

Notes, labels and title blocks, typed right on the drawing.

Paragraph text (MTEXT)

MTEXT (`T`, `MT`): pick two corners and an editor opens in that box. The text wraps at the box's width. `Enter` starts a new line; `Ctrl+Enter`, OK, or a click outside finishes; `Esc` cancels.



Typing a note with MTEXT: the formatting bar is under the text box.

The formatting bar has the text style, font, height, **Bold**, Italic and Underline (`Ctrl+B`, `Ctrl+I`, `Ctrl+U`), color, alignment, and a Symbols menu (degree, plus-minus, diameter...). Font, height, bold, italic, underline and color change only the selected characters. Drag the grips on the top corners of a selected MTEXT to change its width.

Single-line text (TEXT)

TEXT (`DT`): pick the start point, the height and the angle, then type; each line is its own object. `Enter` on an empty line ends. Double-click any text (or **TEXTEDIT**, `ED`) to edit it.

Text styles and fonts

STYLE (`ST`, Format > Text Style) sets a style's font, bold, italic, fixed height, width factor and oblique angle. Any installed font works, and AutoCAD `.shx` font names are shown with a similar font. To add fonts, put `.ttf` or `.otf` files in `~/.config/fancypencil/fonts`; they are installed the next time Fancy Pencil starts. Drawings that use Century Gothic show it when it is installed.

Spelling and case

SPELL (`SP`) checks text, MTEXT, attributes and dimension text, with Change, Change All, Ignore and Add to Dictionary. While typing, unknown words are underlined in red; right-click one for suggestions. `Shift+F3` in the editor changes the case of the selected words.

Text into outlines (TXTEXP)

TXTEXP (Express > Explode Text) turns text into closed polylines that trace each letter, in the same font you see on the screen, with the capitals as tall as the text height. Use it for lettering to cut, engrave or extrude: **EXTRUDE** the outlines to make raised letters. **EXPLODE** (`X`) does the same to TEXT; on MTEXT it first makes one TEXT for each line, so explode twice to get outlines. The outlines keep the text's layer and color. Undo puts the text back.



Above, ordinary text; below, the same text after TXTEXP.

CHAPTER 10

Dimensions and Leaders

Measurements and callouts that look like AutoCAD's.

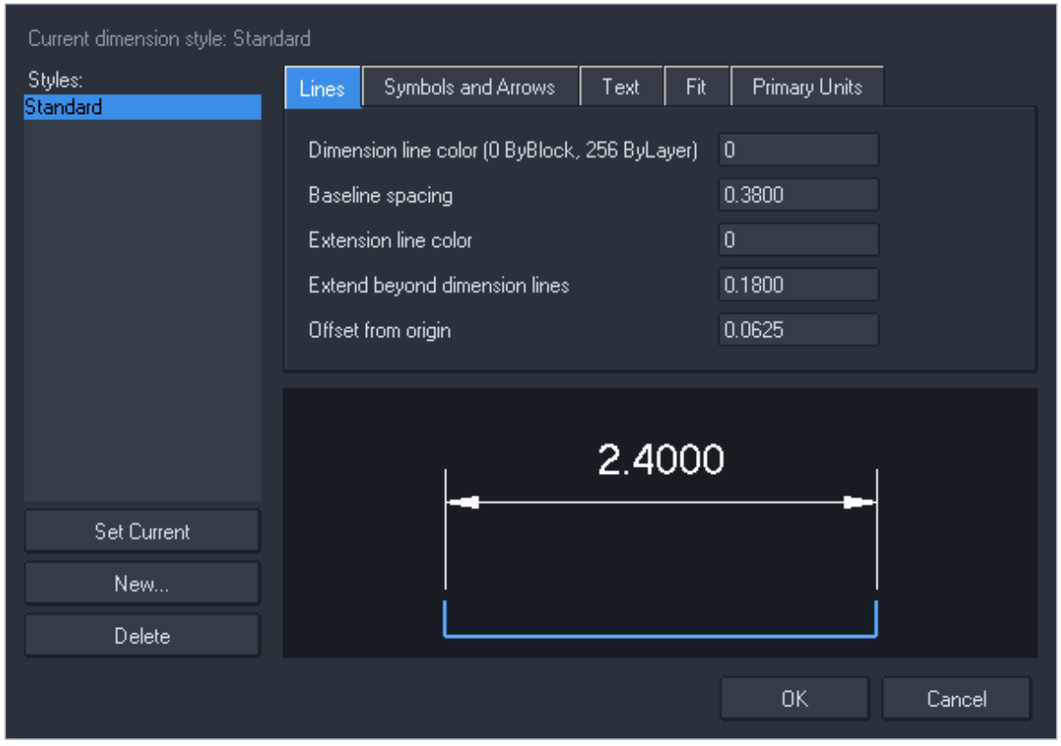
Dimension commands

DIMLINEAR (`DLI`)	Horizontal or vertical: two extension-line origins (or `Enter` and pick an object), then where the dimension line goes.
DIMALIGNED (`DAL`)	Parallel to the two points.
DIMRADIUS (`DRA`)	A circle's or arc's radius.
DIMDIAMETER (`DDI`)	A circle's diameter.
DIMANGULAR (`DAN`)	The angle between two lines, or of an arc.
DIMCONTINUE (`DCO`)	A chain of dimensions from the last one.
DIMBASELINE (`DBA`)	Dimensions stacked from the same first point.

Bring a new dimension line near an existing parallel dimension and it snaps to the same line, so rows of dimensions line up. Grips at the arrows move the dimension line.

Dimension styles

DIMSTYLE (`D`) opens the Dimension Style Manager: styles on the left (Set Current, New, Delete), and tabs for Lines, Symbols and Arrows, Text, Fit and Primary Units, with a live preview. Arrowheads include closed filled, open, dot, architectural tick and oblique. Changing a style updates every dimension that uses it.



The Dimension Style Manager.

Leaders and multileaders

LEADER (`LE`): click the arrow tip, then the points of the line, press `Enter` and type the note. **Spline** or **STraight**, **Arrow** or **None**, and **NOText** are options at every point prompt. **MLEADER** (`MLD`) makes AutoCAD's newer multileader, whose arrow, line, landing and note are one object. Arrow size follows `DIMASZ` and the dimension style.

CHAPTER 11

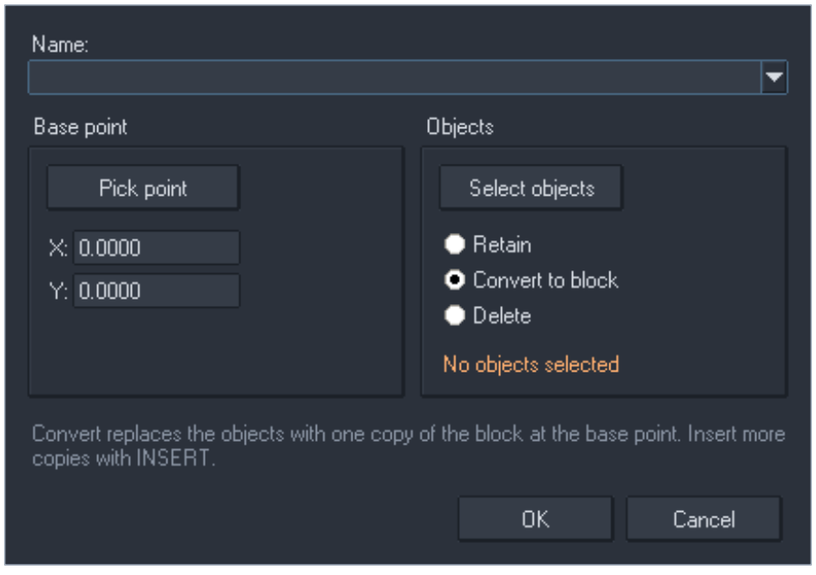
Blocks

Draw a part once, then use it again and again.

A block is a group of objects saved under a name - a bolt, a door, a title block. Every copy (insert) of it changes when the block is edited.

Making a block

BLOCK (`B`, Draw > Block: Make): type a name, **Pick point** for the base point (the point you will insert it by), **Select objects**, and choose whether the objects are retained, converted to the block, or deleted. Objects selected beforehand are used at once.



The Block Definition dialog.

Inserting, editing and renaming

- **INSERT** (`I`): pick the block, the insertion point, scale and rotation.
- **BEDIT** (`BE`, or right-click a block > Edit Block...): the block opens in its own Block Editor tab with its base point at the red cross. Change it, then **BSAVE** (or `Ctrl+S`) and every copy updates. **BCLOSE** finishes.
- **RENAME** renames a block, layer, text style or dimension style everywhere it is used.
- **EXPLODE** (`X`) turns one insert back into separate objects; **BURST** keeps attribute values as text.
- Right-click a selection > **Create Block from Selection...**; Edit > **Paste as Block** (`Ctrl+Shift+V`).

A block made from objects at different heights keeps its 3D shape when inserted.

CHAPTER 12

Views and 3D Navigation

Zoom, pan, turn the drawing around and look at it from any side.

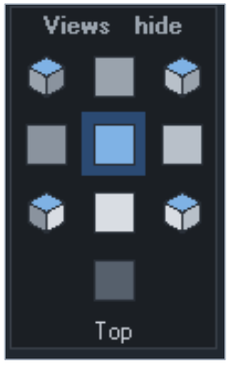
Zoom and pan

- Turn the mouse **wheel** to zoom at the cursor; **middle-drag** to pan; **middle double-click** to see everything.
- **ZOOM** (Z): Extents (E), All, Object (0: the objects you pick), Window (W), Previous (P), Center, or a scale factor. **PAN** (P).

When the drawing looks like a speck

If Zoom Extents shows your work as a dot, something in the drawing is far away from the rest, or far too big: a 3D move that went a hundred million units up, or a circle miles across. **STRAYS** (Tools > Find Stray Objects) finds them. It lists the drawing's groups of objects, with their size and where they are, and any very large objects. Pick a group to select it and zoom to it; then **MOVE** puts it back, or **ERASE** removes it. To keep a large object but out of the way, put it on a layer of its own and freeze that layer.

The view picker and standard views



The view picker.

The **view picker** at the top right shows a cube for each standard view: Top in the middle, the Front, Back, Left and Right views around it, the four isometric views in the corners, and Bottom underneath. Click one to switch. **-VIEW** and **PLAN** do the same at the command line.

In every view you can draw and edit: Front, Back, Left, Right and Bottom each work like an AutoCAD UCS, with X to the right, Y up and Z toward you. A rectangle drawn in the Front view stands upright.

Orbit

3DORBIT (`3DO`)	Drag to turn the view around the target; up stays up.
3DFORBIT	Free orbit: turn any way inside the green circle, roll outside it.
3DCORBIT	Drag and let go: the view keeps turning. Click to stop.
3DDISTANCE, **3DSWIVEL**	Move the camera closer or away; turn the camera like turning your head.
`Shift` + middle-drag	Orbit at any time, even in the middle of a command.

Visual styles

VSCURRENT (`VS`, View > Visual Styles): **Wireframe** shows every edge, **Hidden** hides what is behind solids, **Shaded** fills and lights them, and **Shaded with Edges** adds the edges. Plots follow the visual style.

Working planes

UCS sets any plane as the working plane: an object's or face's plane, three points, a new origin, or turned about X, Y or Z. **DRAFTPLANE** (`F6`) changes the plane you draw on in a 3D view without changing the view: Ground, Front, Right, three points, or an object's plane.

Viewports on the Model tab

VPORTS splits the drawing area into 2, 3 or 4 viewports, each with its own view and style; **4 > Standard** gives Top, SE Isometric, Front and Right, lined up. Click a viewport to make it active (blue border); **Single** goes back to one.

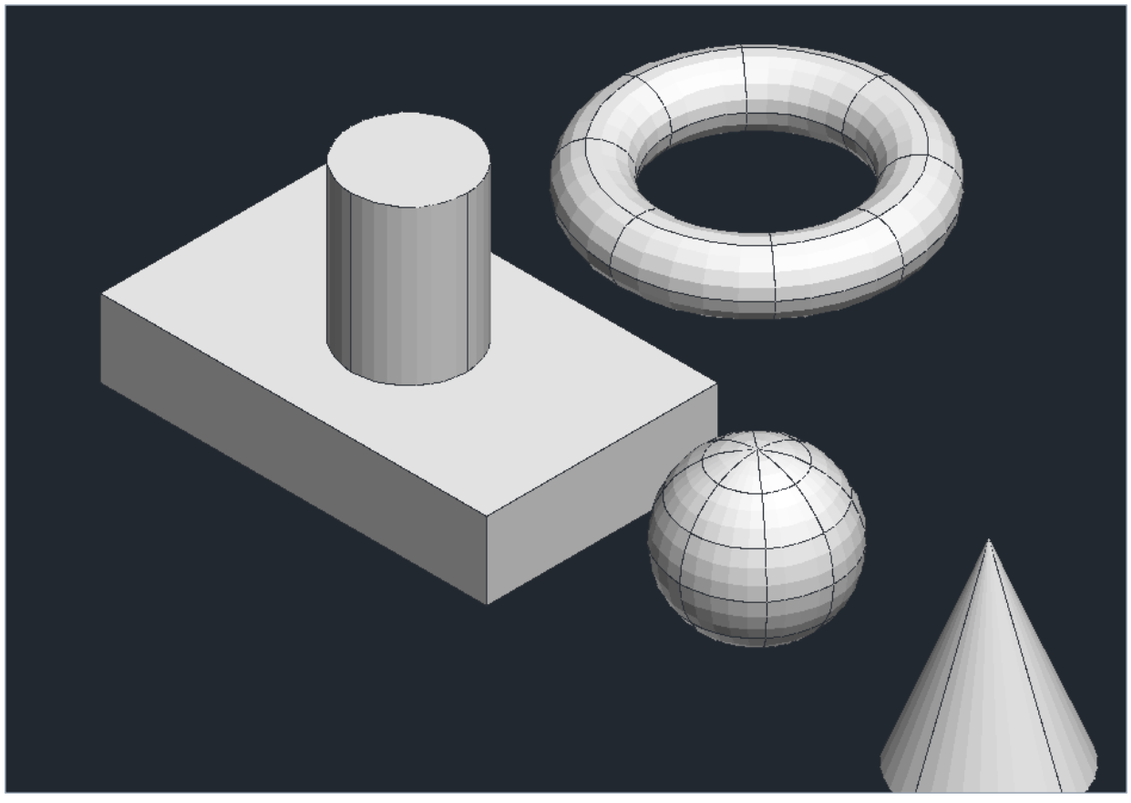
Isometric drafting

ISODRAFT (the ISO button) turns on AutoCAD's 2D isometric mode: an isometric grid, and crosshairs and Ortho that follow the isoplane. `F5` switches between the Left, Top and Right isoplanes; ELLIPSE Isocircle draws a circle as it looks on the current one.

CHAPTER 13

3D Solids

Build solid models, join and cut them, and print them in 3D.



Box, cylinder, sphere, cone and torus in the SE Isometric view, Shaded with Edges.

Basic solids

BOX, **WEDGE**, **CYLINDER**, **CONE**, **SPHERE** and **TORUS** (Solids menu) are drawn in the working plane, so a box drawn in the Front view stands upright. Heights may be negative.

Solids from shapes

EXTRUDE	Pulls a closed shape (circle, closed polyline, ellipse, closed spline) into a solid: by a height, a direction, or along a path.
REVOLVE	Turns a shape about an axis by any angle.
SWEEP	Carries a shape along a path - a line, arc, polyline, spline or helix.

LOFT	A solid through cross sections picked in order (a circle to a square, say).
PRESSPULL	Click inside a closed area, or on a flat face of a solid, and give a height. On a face, a positive height adds; a negative one cuts in.

Joining and cutting

UNION joins solids into one. **SUBTRACT** removes some solids from others: pick what to keep, `Enter`, then what to remove. **INTERSECT** keeps only the shared part. **SLICE** cuts with a plane, keeping one side or both, and closes the cut with a flat face (left open where an imported mesh has gaps). All four are on the right-click menu under **Solid Editing** when solids are selected.

Exact solids

Solids are **exact**: a cylinder is a true circle all the way round, however far you zoom in, and its volume is exactly pi r^2 h. The geometry is done by **Open CASCADE** (the engine inside FreeCAD), which comes with Fancy Pencil. MOVE, ROTATE, SCALE and MIRROR keep solids exact, and so do UNION, SUBTRACT, INTERSECT and SLICE. The exact shape is saved in the DXF with the solid; AutoCAD sees its facets.

FILLETEDGE	Rounds edges: pick them, then `Enter`. `R` sets the radius.
CHAMFEREDGE	Bevels edges the same way. `D` sets the distance.
CONVTOSOLID	Makes a facet solid (an STL, an older drawing, a mesh from a DWG) exact, keeping its flat faces.
STEPOUT, **STEPIN**	Save solids as a STEP file (.step) for other CAD programs and machine shops, or bring one in. STEP is in millimetres: Fancy Pencil converts.
SOLIDENGINE	Says whether solids are exact. `SOLIDMODE` 0 makes facet solids instead.

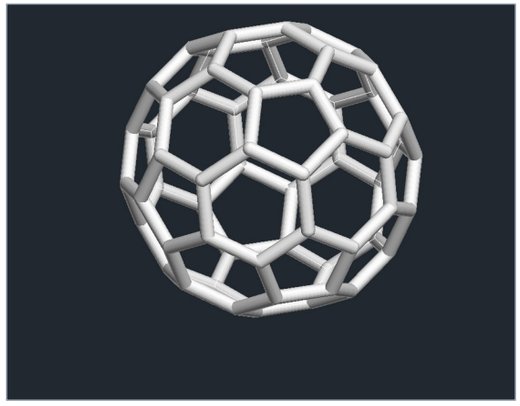
Sweeps, lofts, Presspull and the Platonic solids are still made of facets: CONVTOSOLID makes them exact.

Taking a solid apart

EXPLODE (`X`) splits a solid into its faces, as AutoCAD does: each flat face becomes a **region** and each curved one a **surface**. A box gives six regions; a cylinder two round regions and one surface. The faces keep the solid's layer and color, so you can move one aside, or erase one to look inside. Explode a region or surface again to get its edges as lines. Undo puts the solid back.

Platonic solids, the soccer ball and struts

TETRAHEDRON, **OCTAHEDRON**, **DODECAHEDRON**, **ICOSAHEDRON** and **SOCCERBALL** (Solids > Platonic Solids & Soccer Ball): give the centre and the edge length. Type `W` for a wireframe (3D lines you can snap to), or `F` for a **Frame** of solid struts along every edge.



A soccer ball frame: round struts with a ball at every corner, Shaded.

STRUTS (`WIRESOLID`) does the same for any wireframe you draw: choose the section - **Rectangle**, **Polygon**, **Circle**, or **PLine** (a closed polyline you drew) - and whether to put a ball at every corner. Curved polylines, arcs and circles get one strut that follows the curve. **Ends** sets how far the struts stop short of the wire ends (negative: past them).

Measuring and printing

MASSPROP reports volume, centroid and moments. **3DPRINT** saves solids as STL, 3MF or OBJ for a 3D printer's slicer, checking that each is watertight; **STLIN** brings in STL or OBJ models.

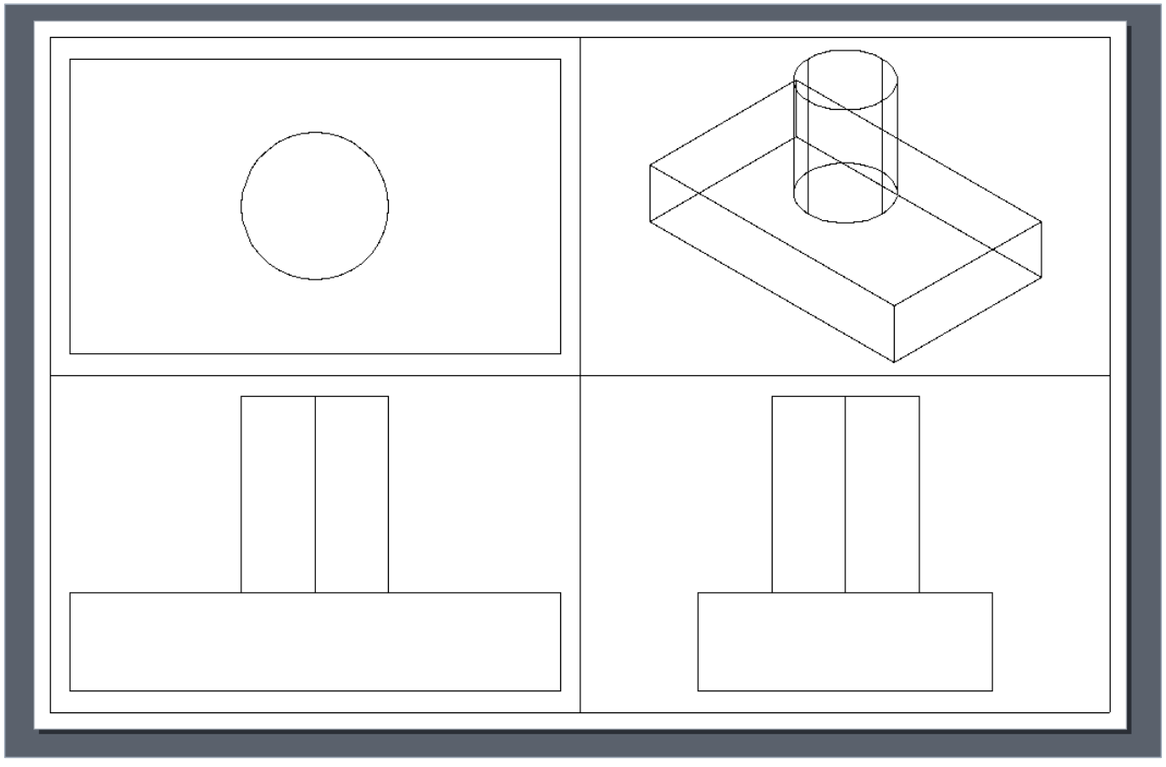
CHAPTER 14

Layouts and Plotting

Sheets of paper, viewports at scale, and printing to PDF or a printer.

Layouts

Below the drawing are the **Model** tab and layout tabs (**Layout1**, and more with **+**). A layout is a sheet of paper at its real size, with a dashed line for the printable area. Draw the border, title block and notes on it; they belong to that sheet.

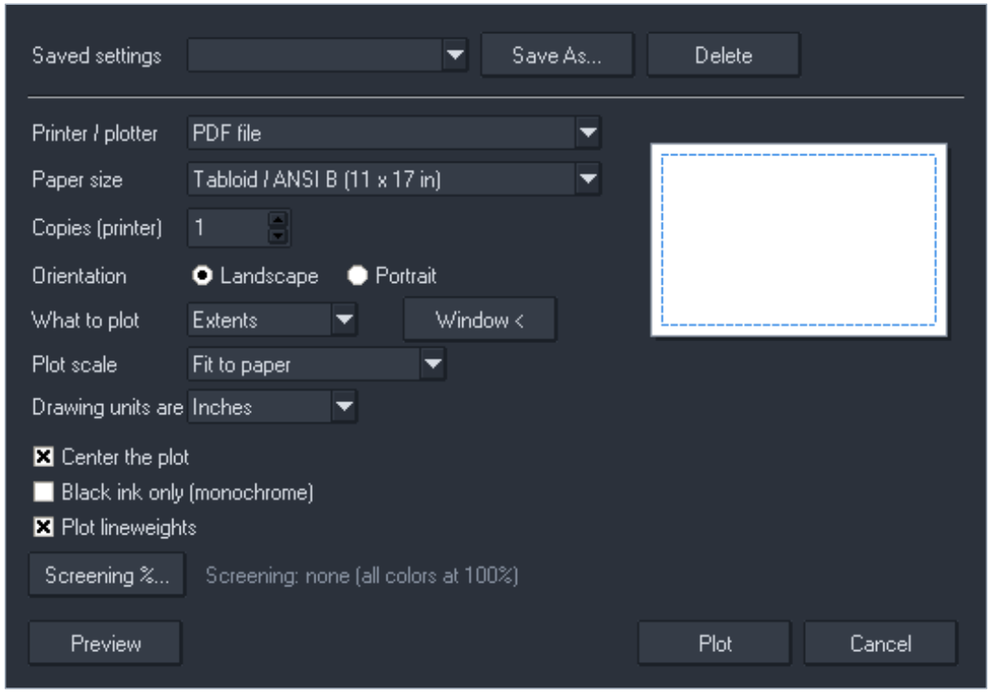


A layout with four viewports (MVIEW 4, Standard): Top, Front, Right and SE Isometric.

Viewports on a layout

- **MVIEW** (`MV`): two corners, **Fit**, or **2**, **3** or **4** at once; **4 > Standard** makes Top, Front, Right and SE Isometric, lined up.
- **Double-click inside** a viewport (or `MSPACE`) to work through it: zoom, pan, change the view. **Double-click outside** (or `PSPACE`) to go back to the sheet.
- **VPSCALE**: 1:2, 1/4"=1'-0", 1:100, Fit... Properties shows and changes a selected viewport's scale, view and lock.
- **PAGESETUP** sets the sheet's paper size and orientation; right-click a layout tab to rename, copy or delete it.

The Plot dialog



PLOT (Ctrl+P).

- **Printer/plotter**: "PDF file" or any printer your computer knows.
- **Paper size**: Letter, Legal, Tabloid, ANSI C-E, ARCH A-E, ISO A4-A0; portrait or landscape.
- **What to plot**: Display, Extents, Limits or Window. On a layout, the sheet at 1:1.
- **Plot scale**: Fit to paper, 1:1, 1:2 ... or architectural scales such as `1/4" = 1'-0"`.
- **Black ink only**, **Plot lineweights**, and **Screening %** to print colors lighter.
- **Saved settings**: Save As keeps everything under a name. Your last settings come back next time.
- **Preview** shows the page first.

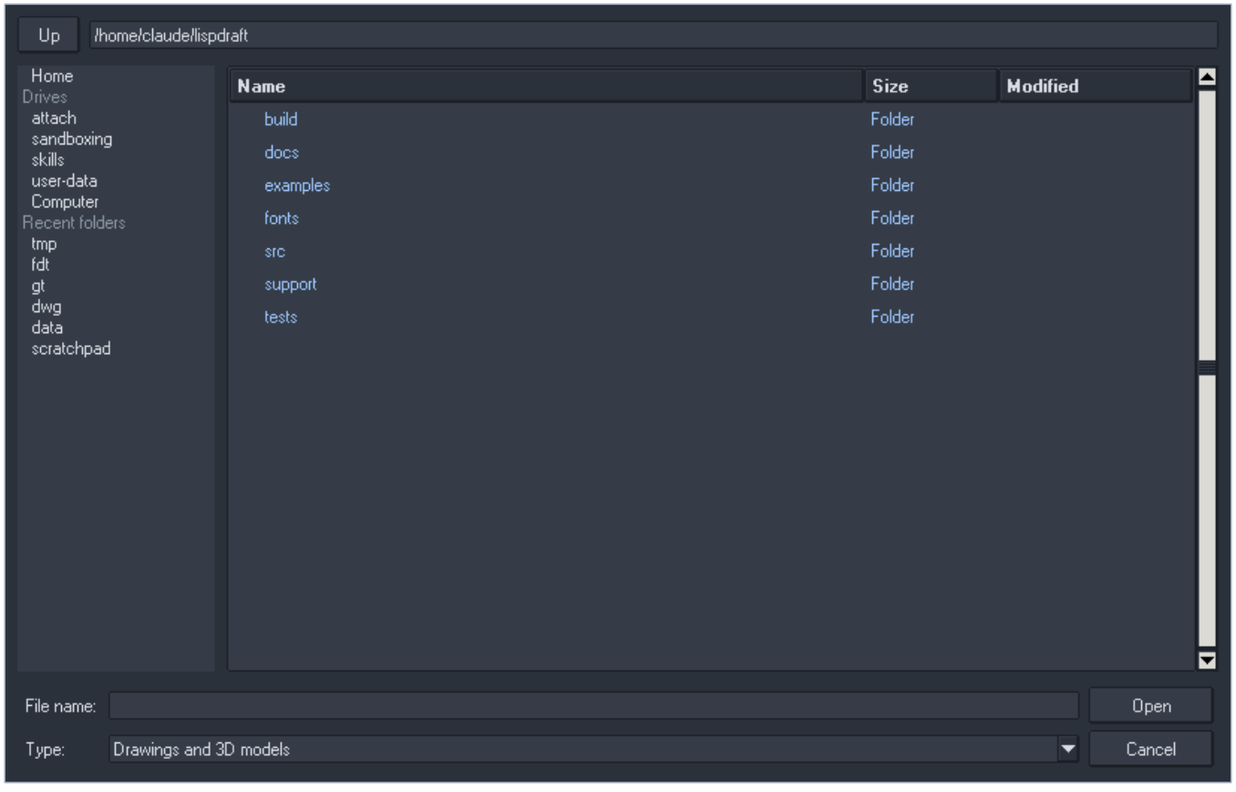
CHAPTER 15

Files: Opening, Saving and Safety Copies

DXF, DWG, pictures and PDFs - and what happens if the power goes.

Opening and saving

OPEN (`Ctrl+O`) shows Fancy Pencil's own file chooser: your folders, drives and recent folders on the left, the files with sizes and dates on the right. Type a path at the top and press `Enter` to go straight there. **File > Recent Files** lists the last 12 drawings.



The Open dialog.

QSAVE (`Ctrl+S`) saves; **SAVEAS** saves under a new name. Drawings are saved as AutoCAD 2000 DXF, which every AutoCAD since 2000 opens. **SAVEASR12** writes old R12 DXF. Saving over a drawing keeps the previous version beside it as a `.bak` file.

DWG files

With ODA File Converter installed (Chapter 2), OPEN shows DWG files too, and Save writes them back as DWG (AutoCAD 2018 format). Without it, saving a DWG asks for a DXF name instead and never touches the DWG.

A DWG (or DXF) brings in its **raster images**, **PDF, DWF and DGN underlays** (looked for beside the drawing), **regions** and **3D solids**, **tables**, and its **paper-space layouts** with their viewports. A table comes in as a block showing its grid and text: edit it with **EXPLODE** and ordinary text editing. Anything Fancy Pencil can't keep is listed when the drawing opens.

Safety copies and recovery

Every 10 minutes, each drawing with unsaved changes is copied to a safe place; your own file is not touched. If Fancy Pencil stops unexpectedly, the next start offers to open the copies. **File > Recover Safety Copies** opens them at any time, and `SAVETIME` sets the minutes between copies.

Pictures, PDF pages and underlays

IMAGEATTACH places a JPEG, PNG, TIFF, GIF, BMP or WebP picture to trace over; **PDFATTACH** places a page of a PDF. Click the lower-left corner and give the width. **IMAGE** (the Image Manager) lists them, and Detach, Reload, Fade and Change Path. **DWFATTACH** and **DGNATTACH** attach DWF and MicroStation drawings as underlays. Keep the picture files with the drawing.

3D printing files

3DPRINT writes STL, 3MF or OBJ; **STLIN** (or File > Open on an STL or OBJ) reads them.

CHAPTER 16

Toolbars and Your Settings

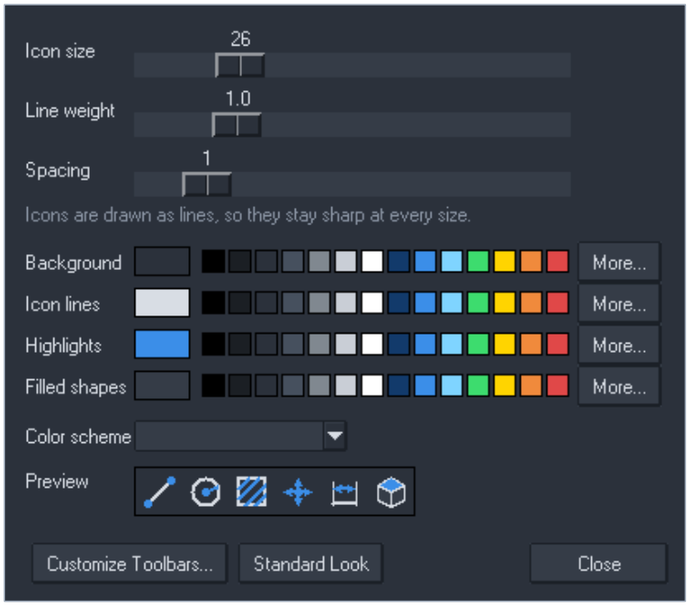
Arrange the screen the way you like it.

Turning toolbars on and moving them

- **View > Toolbars**, or right-click any toolbar, to turn toolbars on and off. **Reset Toolbar Layout** brings back the standard one.
- Drag a toolbar by its dotted grip. Drop it near an edge to dock it there - top, bottom, left or right, even just outside the window's edge - or in the middle of the drawing to float it.
- A floating toolbar can also be dragged by its title bar: let go of it at an edge of the window and it docks there.
- **Clean Screen** (`Ctrl+0`, or View > Clean Screen) hides the ribbon and every toolbar so the drawing gets the whole window; `Ctrl+0` again brings them back.
- **Bring Floating Toolbars Back** (View > Toolbars) gathers every floating toolbar into the top left of the drawing, if any have wandered off or ended up behind other windows.

Toolbar Options

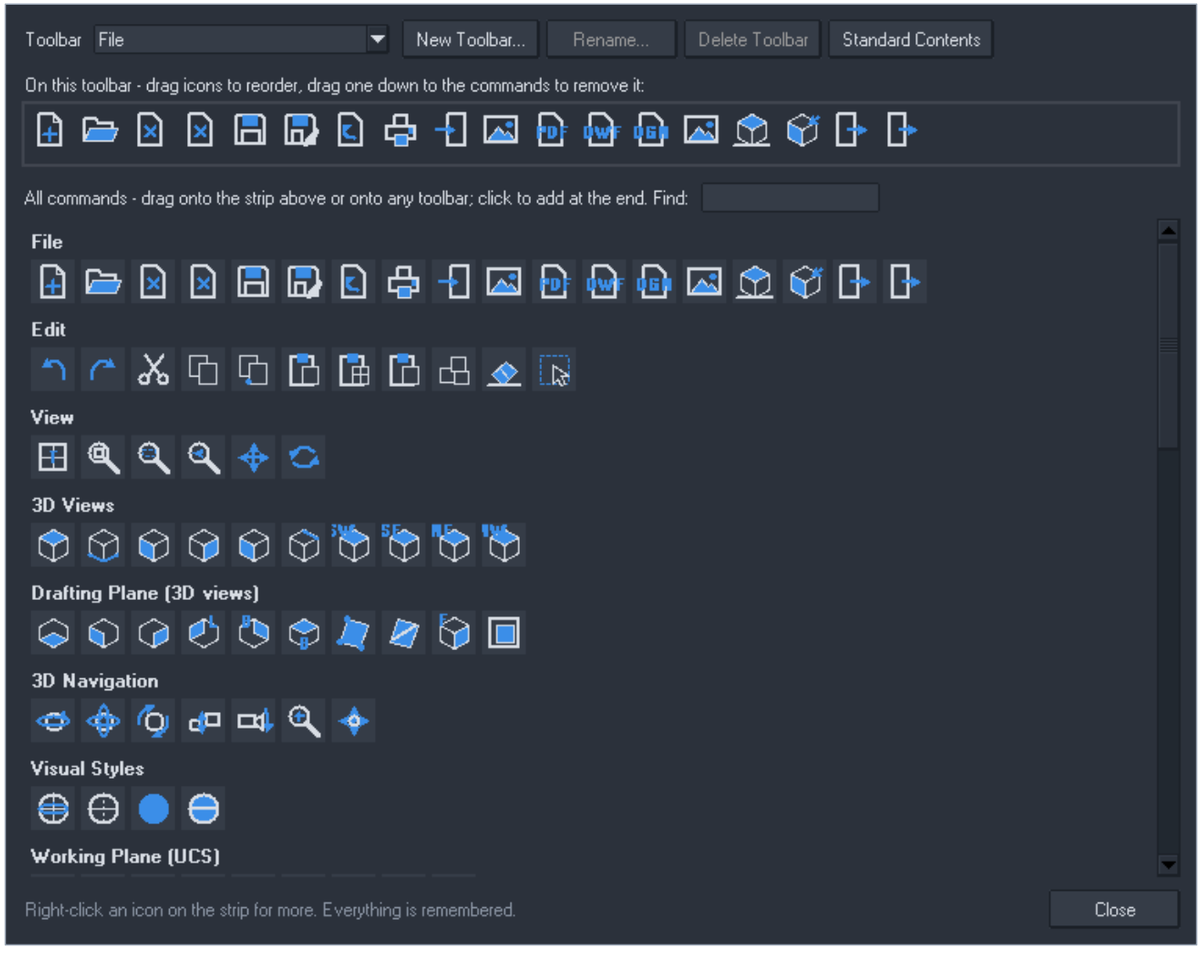
Toolbar Options (View > Toolbars, or `TOOLBAROPTIONS`) stays open while you work: sliders for icon size (16 to 64 pixels), line weight and spacing, colors for the background, icon lines, highlights and filled shapes, and ready-made color schemes. **Standard Look** puts everything back.



Toolbar Options.

Customize Toolbars

Customize Toolbars is all drag and drop. At the top, the toolbar you are editing (or **New Toolbar...** for your own); below, every command's icon by menu, with a Find box. Drag a command onto the strip - or straight onto any toolbar on the screen. Drag icons along a toolbar to reorder them, or down onto the commands to remove them. At other times, `Shift`+drag moves a toolbar button.



Customize Toolbars.

Command aliases

Tools > Command Aliases > Edit Aliases... (`ALIASEDIT`) changes the short names: `L` for LINE, your own for your AutoLISP commands. Your aliases are kept in `~/.config/fancypencil/fancypencil.pgp`, in AutoCAD's acad.pgp format.

Cursor, selection and background

OPTIONS (`OP`, Tools > Cursor, Selection & Background): crosshair size and color, pickbox, snap marker and grip sizes, rollover tips, and the drawing background (dark, black, gray, paper, white or any color). These settings are yours: they apply in every drawing and are remembered.

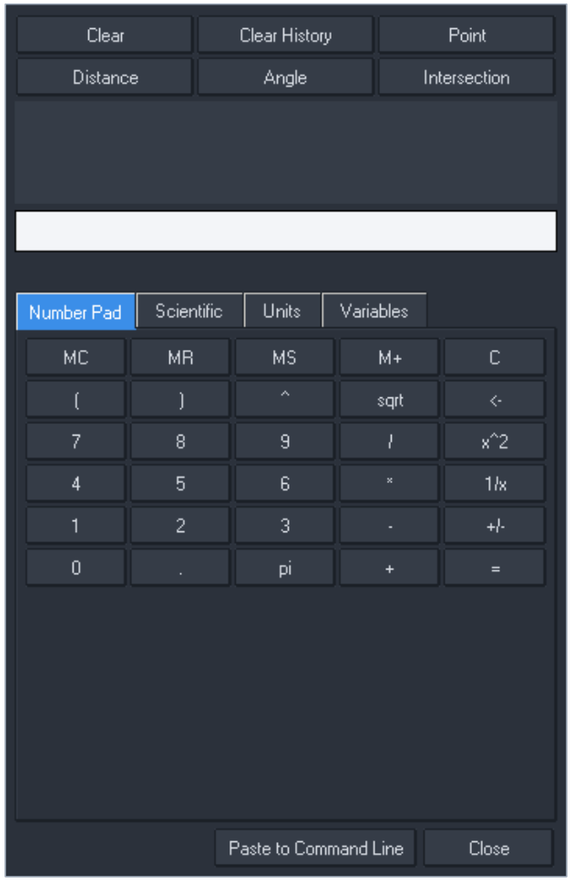
CHAPTER 17

Tools

Calculating, units, limits and the command history.

QuickCalc

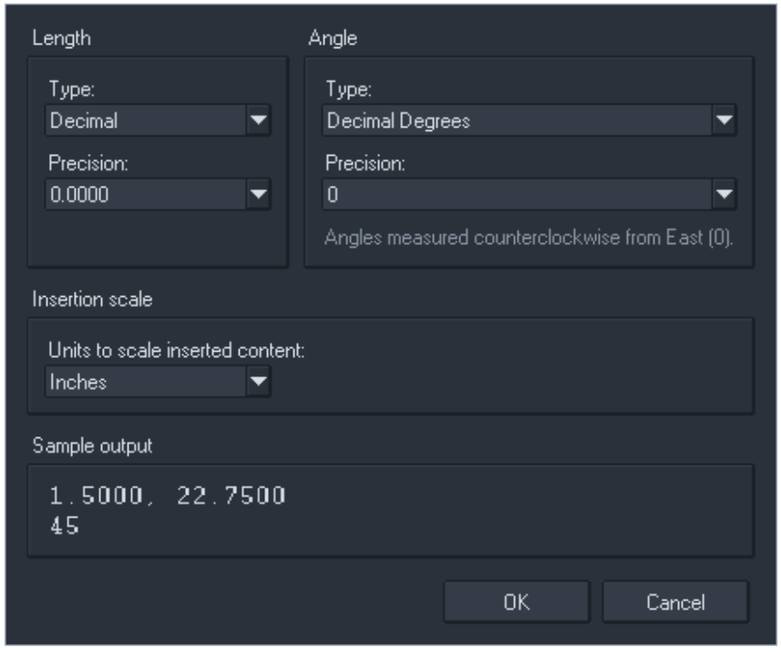
QUICKCALC (`QC`, `Ctrl+8`) opens a calculator beside the drawing. Type an expression and press `Enter`: `5'6"*2`, `sqrt(2)*3`, `dist(end,end)/2`. It has number and scientific keys, unit conversion and variables, and buttons that read a point, distance or angle from the drawing. In the middle of a command, type `'QC` at a prompt, work out the value, and **Apply** gives it to the command.



QuickCalc.

Units

UNITS (Format > Units): how lengths are shown - Architectural (`12'-6 1/2"`), Decimal, Engineering, Fractional or Scientific - with their precision; angles in degrees, deg/min/sec, grads, radians or surveyor's units. The sample shows the result as you choose.



Drawing Units.

Limits, point style and settings

- **LIMITS** (Format > Drawing Limits): the drawing's size, from corners or a sheet size and scale.
- **DDPTYPE** (Format > Point Style): the 20 point symbols, and their size.
- **SETVAR**, or type any system variable's name, to see or change it.

Command history

HISTORY (F2) shows everything from the command line for the whole session, with Find, Copy All and Save As. Tick **Keep a log file** to have it saved every day.

CHAPTER 18

AutoLISP

Run AutoCAD's programming language, and your own routines.

Fancy Pencil's AutoLISP follows AutoCAD's rules: `(defun c:name (args / locals) ...)` makes a command, symbols aren't case-sensitive, `(/ 7 2)` is 3, and `*error*` handlers work. About 250 functions are built in.

Loading routines

- **APPLOAD** (`AP`, Tools > Load AutoLISP...) loads a `.lsp` file.
- `(load "myroutine.lsp")` at the command line does the same.
- Files named `acaddoc.lsp` or `fancypencil.lsp` in the current folder, and `~/.fancypencil.lsp`, load at startup, and `S::STARTUP` runs if defined.
- Give a routine an alias in your alias file: `HH, *HELLO` runs `C:HELLO`.

An example

```
(defun c:box2 (/ p)
 (setq p (getpoint "\nCorner: "))
 (command "rectang" p (polar p 0 4) )
 (princ))
```

Type that at the command line (or put it in a file and load it), then `BOX2`.

What is built in

Core	car/cdr, cons, list, append, nth, member, assoc, mapcar, apply, lambda, foreach, repeat, while, cond, if, setq, defun
Math	+ - * / rem abs fix float max min sqrt expt exp log sin cos atan logand
Strings	strcat strlen substr strcase itoa atoi atof rtos distof angtos chr ascii wcmatch, vl-string-*
Geometry	distance angle polar inters trans osnap textbox
Input	getpoint getcorner getdist getangle getreal getint getstring getkword initget entsel grread getfiled alert
Drawing	entget entmod entmake entdel entlast entnext handent tblsearch tblnext ssget ssadd sslength ssname sssetfirst
Commands	command, command-s, vl-cmdf, with pause

Visual LISP	vl-sort vl-remove-if vl-position vl-some vl-every vl-catch-all-apply ...
Express Tools	acet-error-init, acet-str-*, acet-ss-*, acet-geom-*, acet-pline-make ...

`ssget` supports interactive selection, "X", "W", "C", "WP", "CP", "F", "L", "P", "I" and ":S" with filter lists, wildcards and `-4` operators. Not available: ActiveX (`vla-*`, `vlax-*`) and DCL dialogs.

Examples that come with Fancy Pencil

`/opt/fancypencil/examples/demo-plate.lsp` draws the dimensioned plate on page 1 (load it and type `PLATE`). `tools.lsp` has TLEN (total length), BCIRC (bolt circle), NUM (numbering) and CL (centre marks).

CHAPTER 19

Keyboard Shortcuts

The keys that save the most time.

Key	Command	What it does
`Enter` `Space`		Finish typing; repeat the last command at an empty prompt.
`Esc`		Cancel the command; clear the selection.
`Delete`	ERASE	Erase the selected objects.
`Ctrl+N`	NEW	New drawing.
`Ctrl+O`	OPEN	Open a drawing.
`Ctrl+S`	QSAVE	Save.
`Ctrl+P`	PLOT	Plot.
`Ctrl+Z` / `Ctrl+Y`	U / REDO	Undo / redo.
`Ctrl+C` / `Ctrl+X`	COPYCLIP / CUTCLIP	Copy / cut to the clipboard.
`Ctrl+V`	PASTECLIP	Paste at a point.
`Ctrl+Shift+C`	COPYBASE	Copy with a base point.
`Ctrl+Shift+V`	PASTEBLOCK	Paste as a block.
`Ctrl+A`	SELECTALL	Select everything.
`Ctrl+0`	CLEANSCREENON	Clean Screen: hide the ribbon and toolbars (again to show).
`Ctrl+1`	PROPERTIES	Properties palette.
`Ctrl+8`	QUICKCALC	Calculator.
`Ctrl+Tab`		Next drawing tab (`Ctrl+Shift+Tab`: previous).
`Ctrl+F4`	CLOSE	Close the drawing.
`F1`	MANUAL	Open this manual.
`F2`	HISTORY	Command history.
`F3`	OSNAP	Object snaps on/off.
`F5`	ISOPLANE	Next isoplane (isometric drafting).
`F6`	DRAFTPLANE	Next drafting plane in 3D views.

Key	Command	What it does
`F7`	GRID	Grid on/off.
`F8`	ORTHO	Ortho on/off.
`F9`	SNAP	Snap on/off.
`F10`	POLAR	Polar tracking on/off.
`F12`	DYN	Dynamic input (prompts at the cursor) on/off.
Wheel		Zoom at the cursor.
Middle-drag	PAN	Pan.
Middle double-click	ZOOM E	Zoom to everything.
`Shift`+middle-drag	3DORBIT	Orbit, at any time.
`Down` arrow		During a command: a menu of its options. At the Command prompt: earlier commands (with `Up`).
Right-click		Menu at the Command prompt; Enter during a command.
`Shift`+right-click		One-time object snap menu.

CHAPTER 20

Command Reference

Every command, A to Z, with its aliases (the short names you can type instead).

3D

3DCORBIT Continuous orbit: drag and let go, and the view keeps turning.

3DDISTANCE Moves the camera closer to the objects or farther away (drag up and down).

3DFORBIT Free orbit: turns the view in any direction, without keeping up up.

3DORBIT `3DO, ORBIT` Turns the view around a target point (constrained orbit: up stays up).

3DPAN Pans a 3D view (drag).

3DPRINT `EXPORT3D, PRINT3D` Saves solids as a file for a 3D printer's slicer: STL, 3MF or OBJ.

3DSWIVEL Turns the camera where it stands, like turning your head: the target moves.

3DZOOM Zooms in a 3D view (drag up to zoom in).

A

ABOUT Shows program information.

ALIASEDIT `ALIASES` Edits command aliases.

ALIASFILE `PGP` Opens your command alias file (fancypencil.pgp) in a text editor.

ALIASFOLDER Opens the folder that holds your command alias file.

ALIGN `3DALIGN, AL` Moves, rotates and optionally scales objects so up to three points line up with three others, in 3D.

APPLOAD `AP` Loads an AutoLISP (.lsp) file.

ARC `A` Creates an arc.

AREA `AA` Calculates the area and perimeter of objects or a polygon of points.

ARRAY `AR` Creates multiple copies in a rectangular or polar pattern.

ARRAYPOLAR Polar array.

ARRAYRECT Rectangular array.

ATTACH Attaches a picture or a PDF page to the drawing (choose the file).

B

BACKGROUND `BGCOLOR, BKG` Sets the drawing area's background color.

BCLOSE Closes the Block Editor.

BEDIT `BE, BLOCKEDIT, REFEDIT` Opens a block in the Block Editor to change it.

BLOCK `B, BMAKE` Creates a block definition from selected objects (Block Definition dialog).

-BLOCK Creates a block definition at the command line.

BOX Creates a 3D solid box in the working plane.

BREAK `BR` Breaks the selected object between two points.

BREAKLINE `BREAKL` Express: draws a break-line symbol.

BRINGTOFRONT Draws the selected objects over everything else.

BSAVE Saves the Block Editor's changes into the block.

BURST Express: explodes blocks, turning attribute definitions into text.

C

CAL Calculates an expression (numbers, points, geometry picked in the drawing).

CHAMFER `CHA` Bevels the edges of objects.

CHAMFEREDGE `CHAMFEREDGES` Bevels the edges of exact solids.

CHANGELAYER `CL, LAYCHANGE` Moves selected objects to a layer you choose from a list.

CHPROP `CHANGE` Changes the layer or color of objects.

CIRCLE `C` Creates a circle.

CLEANSCREENOFF Brings the ribbon and toolbars back after Clean Screen (Ctrl+0).

CLEANSCREENON Clean Screen: hides the ribbon and all toolbars so the drawing has the whole window (Ctrl+0).

CLEARRECENT Empties the recent files list.

CLOSE Closes the current drawing.

CLOSEALL Closes all open drawings.

COLOR `COL, DDCOLOR` Sets the color for new objects (the Select Color dialog in the window).

CONE Creates a 3D solid cone (or a cut-off cone with a top radius).

CONVTOSOLID `CONVERTTOSOLID` Turns facet solids (meshes, STL imports, older solids) into exact solids made of their flat faces.

COPY `CO, CP` Copies objects.

COPYBASE Copies objects to the clipboard with a base point you pick (Ctrl+Shift+C).

COPYCLIP Copies selected objects to the clipboard (Ctrl+C); paste them into any drawing.

COPYM Express: copies objects repeatedly, by repeat, divide or measure.

CUTCLIP Copies the selection to the clipboard and erases it (Ctrl+X).

CYLINDER `CYL` Creates a 3D solid cylinder standing on the working plane.

D

DDPTYPE `POINTSTYLE, PTYPE` Point Style dialog: how points are shown and how big.

DESELECT Clears the current selection (grips).

DGNATTACH Attaches a MicroStation DGN file as an underlay.

DIMALIGNED `DAL, DIMALI` Creates an aligned linear dimension.

DIMANGULAR `DAN, DIMANG` Creates an angular dimension.

DIMBASELINE `DBA, DIMBASE` Creates a dimension from the baseline of the last one.

DIMCONTINUE `DCO, DIMCONT` Creates a dimension starting at the second extension line of the last one.

DIMDIAMETER `DDI, DIMDIA` Creates a diameter dimension for a circle or arc.

DIMLINEAR `DIMLIN, DLI` Creates a horizontal or vertical linear dimension.

DIMRADIUS `DIMRAD, DRA` Creates a radius dimension for a circle or arc.

DIMREGEN Redraws every dimension's picture from its points and current style.

DIMSTYLE `D, DDIM, DIMSTY, DST` Creates and restores dimension styles.

DIST `DI` Measures the distance and angle between two points.

DIVIDE `DIV` Places evenly spaced point objects along a line, arc or circle.

DOCCLOSE Closes an open drawing (used by the file tabs' close buttons).

DOCNEXT Switches to the next open drawing (Ctrl+Tab).

DOCPREV Switches to the previous open drawing (Ctrl+Shift+Tab).

DOCSWITCH Makes another open drawing current (used by the file tabs).

DODECAHEDRON `DODECA` Creates a dodecahedron: 12 pentagonal faces.

DRAFTPLANE `DPLANE, DRP` Chooses the plane you draw on in the 3D views, without changing the view: Ground, Front, Right, Left, Back, Bottom, through three points, on an object or a solid's face, facing the screen.

DRAWINGRECOVERY `RECOVERCOPIES` Opens safety copies left by Fancy Pencil stopping unexpectedly.

DRAWORDER `DR` Changes which objects are drawn on top: Front, Back, Above or Under other objects.

DWFATTACH Attaches a DWF or DWFx file as an underlay.

DXFIN `IMPORT` Imports a DXF drawing into the current drawing.

DXFOUT Exports the drawing as DXF (choose AutoCAD 2000 or R12).

E

ELEV `ELEVATION` Sets the elevation and thickness for new objects.

ELLIPSE `EL` Creates an ellipse (and isocircles in isometric drafting).

ERASE `DELETE, E` Removes objects from the drawing.

EXPLODE `X` Breaks a compound object into its component objects.

EXTEND `EX` Extends objects to meet the edges of other objects.

EXTRIM Express: trims every object crossing a cutting edge on one side.

EXTRUDE `EXT` Extrudes closed shapes into solids (open ones into surfaces): by a height, along a direction, or along a path.

F

FILLET `F` Rounds and fillets the edges of objects.

FILLETEDGE `FILLETEDGES` Rounds the edges of exact solids.

FLATTEN Express: sets the Z coordinate of selected objects to zero.

G

GETODA `ODA` Opens a window that downloads and installs the free ODA File Converter (for .dwg files).

GETSEL Express: selects objects by layer and type.

GRID Turns the grid on or off, or sets spacing.

H

HATCH `BH, BHATCH, GRADIENT, H` Fills an enclosed area with a hatch pattern or solid fill.

HATCHEDIT `HE` Changes the pattern, scale or angle of existing hatches.

HATCHTOBACK Sends every hatch and fill behind everything else.

HELIX Creates a helix (a spring or thread path) or a flat spiral.

HELP `?` Lists commands.

HIDE `HI` Shows the drawing with hidden lines removed (visual style Hidden).

HISTORY `HIS, TEXTSCR` Shows the whole record of this session: every command, answer and message.

HPSET Sets the hatch pattern, scale and angle used by HATCH.

I

ICOSAHEDRON `ICOSA` Creates an icosahedron: 20 triangular faces.

ID Displays the coordinates of a location.

IMAGE `EXTERNALREFERENCES, IM, IMAGEMANAGER, XR, XREF` The Image Manager: the pictures and PDF pages attached to the drawing.

IMAGEATTACH `IAT` Attaches a picture (PNG, JPEG, TIFF, GIF, BMP) to the drawing.

INSERT `DDINSERT, I` Inserts a block or a DXF drawing.

INTERSECT `IN` Keeps only the part that solids share.

ISODRAFT Turns isometric drafting on (choosing an isoplane) or off.

ISOPLANE Chooses the isoplane for isometric drafting (Left, Top or Right).

J

JOIN `J` Joins lines, arcs and open polylines end to end into one polyline.

K

KEEPPROFILES Keeps (On) or removes (Off) the circles, polylines and paths that EXTRUDE, REVOLVE, SWEEP and LOFT use (DELOBJ).

L

LAYCUR Express: moves selected objects to the current layer.

LAYDEL Express: deletes a layer and every object on it.

LAYER `DDLMODES, LA, LAYERP` Manages layers.

LAYFRZ Express: freezes the layer of a selected object.

LAYISO Express: isolates the layers of selected objects (turns the others off).

LAYLCK Express: locks the layer of a selected object.

LAYMCH `LMH, MATCHLAYER` Changes selected objects to the layer of a destination object (Match Layer).

LAYMCUR `LAYERMAKECURRENT, LMC` Makes the layer of a selected object current.

LAYMRG Express: merges layers into a target layer.

LAYOFF Express: turns off the layer of a selected object.

LAYON Express: turns all layers on.

LAYOUT `LO` Makes, copies, deletes, renames and shows layouts (paper space sheets).

LAYOUTCOPY Copies a layout from its tab.

LAYOUTDELETE Deletes a layout from its tab.

LAYOUTMOVE Moves a layout tab one place left or right.

LAYOUTNEW Adds a layout (the + after the layout tabs).

LAYOUTPAGESETUP Page setup for a layout from its tab.

LAYOUTRENAME Renames a layout from its tab.

LAYOUTTAB Shows a layout tab by its number (0 = Model); used by the tabs under the drawing.

LAYTHW Express: thaws all layers.

LAYULK Express: unlocks the layer of a selected object.

LAYUNISO Express: restores layers turned off by LAYISO.

LEADER `LE, LEAD, QLEADER` Creates a leader: an arrow and a straight or spline line pointing at a note.

LIMITS `DRAWINGLIMITS` Sets the drawing's size (its limits): the area the grid, ZOOM All and Plot Limits use.

LINE `L` Creates straight line segments.

LINETYPE `DDLTYPE, LT, LTYPE` Loads, sets and lists linetypes.

LIST `LI, LS` Displays property data for selected objects.

LOFT Makes a solid (or surface) through cross sections picked in order.

LOGFILEOFF Stops writing the command line to the log file.

LOGFILEON Starts writing everything shown in the command line to a log file.

LTSCALE `LTS` Sets the global linetype scale factor.

LWEIGHT `LINEWEIGHT, LW` Sets the current lineweight and lineweight display.

M

MANUAL `F1, USERMANUAL` Opens the Fancy Pencil Manual (a PDF) in your PDF viewer.

MASSPROP `MASSPROPERTIES` Mass properties of regions (closed flat objects) and solids: area, volume, centroid, moments of inertia ...

MATCHPROP `MA, PAINTER` Copies properties (layer, color, linetype, text style, dimension style, hatch ...) from one object to others.

MIRROR `MI` Creates a mirrored copy of objects.

MLEADER `MLD` Creates a multileader: an arrow, a landing and a note, as one object.

MOCORO Express: moves, copies, rotates and scales with one base point.

MODEL `TILEMODE1` Shows the Model tab.

MOVE `M` Moves objects.

MPEDIT Express Tools: edits several polylines at once.

MSPACE `MS` Works in model space through a layout viewport.

MSPACETOGGLE Switches between paper space and the last viewport (the MODEL/PAPER button).

MTEXT `MT, T` Creates multiline (paragraph) text, typed in place on the drawing.

MULTIPLE-POINT Creates points until Enter.

MVIEW `MV` Makes layout viewports: windows on the sheet that show model space.

N

NEW `QNEW` Creates a new drawing in its own tab.

O

OCTAHEDRON `OCTA` Creates an octahedron: 8 triangular faces.

OFFSET `O` Creates concentric circles, parallel lines, and parallel curves.

OOPS Restores the last erased objects.

OPEN Opens a drawing (DXF, or DWG through a converter), or a 3D model (STL, OBJ) as a solid.

OPENRECENT Opens a drawing from the recent files list (File > Recent Files).

OPTIONS `CURSOR, DDSELECT, OP` Cursor and selection settings: crosshair size and color, pickbox, snap marker, grips, rollover tips.

ORTHO Turns ortho mode on or off.

OSNAP `DDOSNAP, DS, DSETTINGS, OS` Sets running object snap modes (settings panel).

-OSNAP Sets running object snap modes at the command line.

OVERKILL Removes duplicate and overlapping objects.

P

PAGESETUP `PAGESETUPS` Sets the current layout's paper size, orientation and units.

PAN `P` Shifts the view.

PASTEBLOCK Pastes the clipboard as one block (Ctrl+Shift+V).

PASTECLIP Pastes the clipboard at a point you pick (Ctrl+V).

PASTEORIG Pastes the clipboard at the same coordinates it was copied from.

PDFATTACH Attaches a page of a PDF file to the drawing.

PEDIT `PE` Edits polylines.

PLAN Back to the plan (Top) view.

PLINE `PL` Creates a 2D polyline of line and arc segments.

PLOT `PLT, PRINT` Plots the drawing to a PDF file or a printer.

-PLOT Plots at the command line (for scripts).

POINT `PO` Creates a point object.

POLYGON `POL` Creates a regular polygon.

POLYHEDRON `PLATONIC` Creates a Platonic solid (tetrahedron, cube, octahedron, dodecahedron, icosahedron) or a truncated icosahedron (soccer ball).

PRESSPULL `PRESS` Click inside a closed area or on a flat face of a solid (or pick an Edge) and pull it into a solid.

PROPERTIES `CH, DDCHPROP, DDMODIFY, MO, PR, PROPS` Shows the Properties palette.

PSPACE `PS` Goes back to the layout's sheet (paper space) from a viewport.

PURGE `PU` Removes unused layers and block definitions.

Q

QSAVE Saves the current drawing.

QUICKCALC `QC` Opens QuickCalc: a calculator, unit converter, and geometry from the drawing.

QUICKPROPERTIES QP, QUICKPROP Quick Properties: a small panel with the main properties of the selected objects.

QUIT EXIT Exits the program (offering to save each changed drawing).

R

RECTANG REC, RECTANGLE Creates a rectangular polyline.

REDO MREDO Reverses the effects of the previous UNDO.

REGEN R, RE, REDRAW, REDRAWALL, REGENALL Regenerates the display.

REINIT Reloads the command alias (.pgp) file.

RENAME DDRENAME, REN Renames a block, layer, text style or dimension style.

REVCLOUD RC Creates a revision cloud.

REVOLVE REV Revolves shapes about an axis to make solids (or surfaces from open shapes).

REVSURF Creates a revolved surface mesh: a curve spun about an axis (SURFTAB1 x SURFTAB2 faces).

ROTATE RO Rotates objects around a base point.

S

SAVE Saves the drawing.

SAVEAS Saves the drawing under a new name.

SAVEASR12 Saves an AutoCAD R12 DXF (no lineweights or hatches).

SCALE SC Enlarges or reduces objects.

SCRIPT SCR Runs a sequence of commands from a script (.scr) file.

SELECTALL AI_SELALL Selects every visible object.

SELECTSIMILAR SSIM Selects all objects of the same type and layer as the selection.

SENDTOBACK Draws the selected objects under everything else.

SETVAR SET Lists or changes the values of system variables.

SLICE SL Cuts solids with a plane, keeping one side or both.

SNAP SN Sets snap spacing or turns snap on/off.

SOCCERBALL TRUNCICOSA Creates a truncated icosahedron (a soccer ball): 12 pentagons and 20 hexagons.

SOLIDENGINE Says which engine makes 3D solids: exact (Open CASCADE) or facets.

SPELL SP Checks the spelling of text in the drawing.

SPHERE Creates a 3D solid sphere.

SPLINE SPL Draws a smooth curve through (Fit) or controlled by (CV) the points you pick.

SSX Express: selects objects matching a picked object's type, layer and color.

STEPIN IMPORTSTEP, STEPIMPORT Brings the solids from a STEP file (.step, .stp) into the drawing.

STEPOUT EXPORTSTEP, STEPEXPORT Saves solids as a STEP file (.step), the exchange format of 3D CAD programs and machine shops.

STLIN IMPORT3D, OBJIN Brings an STL or OBJ file (from a 3D model or a printing site) in as a solid.

STLOUT Saves solids as an STL file for 3D printing.

STRAYS FINDSTRAYS Finds objects lying far from the rest of the drawing, or far too big for it, to move back or erase.

STRETCH S Stretches objects crossed by a selection window.

STRUTS STRUT, WIRESOLID Turns the lines of a wireframe into solid bars the length of each line, with a rectangle, polygon, circle or drawn-polyline section. Polylines with arcs, arcs, circles and splines get one strut that follows the curve.

STYLE DDSTYLE, ST Creates, modifies, or sets the current text style.

-STYLE Text styles at the command line.

SUBTRACT SU Removes solids from other solids.

SURFTAB1 Number of faces around a revolved surface (REVSURF).

SURFTAB2 Number of faces along a curve of a revolved surface (REVSURF).

SWEEP Sweeps shapes along a path (line, arc, polyline, spline, helix) to make solids or surfaces.

SYSWINDOWS Lists the open drawings.

T

TCASE CHANGECASE Changes the case of text, MText and dimension text: Sentence, lower, UPPER, Title, tOGGLE.

TCOUNT Express: numbers selected text objects.

TETRAHEDRON TETRA Creates a tetrahedron: 4 triangular faces.

TEXT DT, DTEXT Creates single-line text, typed in place on the drawing.

TEXTEDIT DDEDIT, ED, MTEDIT Edits the contents of a text object (in place on the drawing).

TEXTTOFRONT Brings text, dimensions and leaders in front of everything else.

THICKNESS TH Sets the thickness (extrusion height) for new lines, arcs, circles and polylines.

TOOLBAR TB, TO Shows the list of toolbars, to turn them on or off (drag a toolbar by its dotted grip to dock it on any edge or float it).

TOOLBAROPTIONS TBOPTIONS Opens the Toolbar Options window: icon size, line weight, spacing and colors.

TORUS TOR Creates a 3D solid torus (a ring), lying in the working plane.

TRIM TR Trims objects to meet the edges of other objects.

TXT2MTXT T2M Express: combines single-line text into one MTEXT object.

TXTEXP EXPLODETEXT, TEXTEXPLODE Explodes text into closed polylines tracing each letter's outline (Express Tools).

U

U Reverses the most recent operation.

UCS WORKPLANE, WP Sets the working plane: through three points, on an object's plane, at an origin, turned about an axis, facing the view, or back to World.

UNDO Reverses the effect of commands.

UNION UNI Joins solids into one.

UNITS DDUNITS, UN Sets the drawing's length and angle units, precision and insertion units.

V

-VIEW V, VIEW, VIEWS Changes to a standard view: Top, Bottom, Left, Right, Front, Back or an isometric view.

VPORTS VIEWPORTS, VPORT Splits the Model tab into several viewports, each with its own view: Single, Join, 2, 3, 4.

VPSCALE VPS Sets the scale of layout viewports (1:2, 1/4"=1'-0", 2:1, Fit ...).

VSCURRENT `SHADEMODE, VISUALSTYLES, VS` Sets the visual style: Wireframe, Hidden, Shaded or Shaded with Edges.

W

WEDGE `WE` Creates a 3D solid wedge: full height at the first corner, sloping to the opposite side.

X

XLIST Express: lists the properties of an object nested in a block.

Z

ZOOM `Z` Changes the view magnification.

CHAPTER 21

When Something Goes Wrong

Common problems and what to do.

Fancy Pencil doesn't start	Open a terminal and type `fancypencil` to see the message. When started from the menu, messages go to `~/.cache/fancypencil.log`.
It seems stuck	Press `Esc` a few times. If it still doesn't answer, type `pkill -f fancypencil` in a terminal; the next start offers your safety copies.
A .dwg file won't open	Install ODA File Converter (Chapter 2): Help > Get ODA File Converter.
The .deb opens in Archive Manager	Right-click it > Open With... > App Center.
Clicks land in the wrong place	An object snap may be taking the point: press `F3` to turn OSNAP off, or type `NON` before the point.
Lines won't go at an angle	ORTHO is on: press `F8`.
I can't select or change an object	Its layer may be locked or frozen: check the Layer Properties Manager.
The dashes of a linetype don't show	Change LTSCALE: larger for a big drawing, smaller for a small one.
Text looks wrong in AutoCAD	Use fonts AutoCAD has too, or send AutoCAD the font.
A command did the wrong thing	`U` or `Ctrl+Z` undoes it; one U undoes a whole command.
I lost my work in a crash	File > Recover Safety Copies. Saving also keeps the previous version as name.bak.
Out of memory on a big operation	Fancy Pencil stops that command and keeps running. Work on fewer objects at once; U undoes any part done.

Your files and settings

`~/.config/fancypencil/`	Settings, recent files, aliases, toolbars, plot settings, views.
`~/.local/share/fancypencil/autosave/`	Safety copies.
`~/.local/share/fancypencil/logs/`	Command history logs (when turned on).
`~/.cache/fancypencil.log`	Start-up and error messages.
`/opt/fancypencil/`	The program itself (installed by the .deb).